

EMPLOYER



WEST AFRICAN ENERGY SA



SENEGAL NATIONAL ELECTRICITY AGENCY

EPC CONTRACTOR



ÇALIK ENERJİ SAN. VE TİC. A.Ş

CAP DES BICHES COMBINED CYCLE POWER STATION PROJECT (300 MW)

REVISION HISTORY

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REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED

DOCUMENT TITLE/SUBJECT:

TECHNICAL SPECIFICATION FOR LOW VOLTAGE SWITCHGEAR

PROJECT DOCUMENT NO:

<i>Project</i>	<i>Unit</i>	<i>System</i>	<i>Disc</i>	<i>Doc.</i>	<i>Type</i>	<i>Orig.</i>	<i>Serial No</i>	Detailed Information	Rev.	B
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7	Black start /Emergency DG Building electrical equipment layout	SNC-36-UBA-EDF-CEI-001
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10	Water Treatment Plant, Control & Electrical Building Equipment Layout	SNC-36-UGC-EDF-CEI-001
10	HRSG Electrical MCC Room Equipment Layout	SNC-36-UCH-EDF-CEI-001
11	Protection, Synchronization And Metering System Description	SNC-36-BAY-ER-CEI-001
12	LV System Protection & Control Typical Schemes	SNC-36-BL-EDU-CEI-001
13	Senegal Project CIP INITIAL RELEASE	139T0016

DIVISION 1 – GENERAL REQUIREMENTS

DIVISION 1 – SECTION 100

SUMMARY OF WORK

PART 1 – GENERAL

101. PROJECT INFORMATION

- 101.1 Employer : West African Energy
- 101.2 EPC Contractor or Purchaser : CALIK ENERJİ
- 101.3 Purchaser Consultant : Fichtner India
- 101.4 Vendor or Supplier : OEM / Equipment Supplier
- 101.5 Project Name : CAP DES BICHES Combined Cycle Power Station Project
- 101.6 Project Location : Dakar, Senegal.

CAP DES BICHES combined cycle power plant is located in West Africa, Senegal, Dakar district.

102. DESCRIPTION OF THE PROJECT AND GENERAL BACKGROUND

- 102.1 West African Energy (“EMPLOYER”) intends to build a combined cycle power plant at Dakar, Senegal. CALIK ENERJİ (“EPC CONTRACTOR”) has received the order from EMPLOYER to execute the power plant on an Engineering, Procurement, Construction and Commissioning (EPC) basis.
- 102.2 The plant configuration comprise of One (1) combined cycle power block. The block consists of two (2) gas turbine and generators (GTGs), two (2) heat recovery steam generators (HRSGs) & one (1) steam turbine and generator (STG), and Balance of Plant (BoP) equipments. The gas turbine is capable of firing trifuels - Natural gas (Primary fuel), Naphtha (secondary fuel) and Gas Oil (startup fuel) respectively.
- 102.3 Within the scope of the project, the power generated by the new plant will be connected to HV Network to evacuate power to Grid, through new Generator Step up Transformers (GSUT).
- 102.4 The design life of all equipment is 10,000 operations considering projected load transitions, start-ups and shut downs. Equipment shall be designed for cycling service, with daily starts assumed. Equipment shall also be capable of base loaded operation. The plant shall be designed to a life of NOT less than 25

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years.

103. GENERAL

- 103.1 This specification covers the design, manufacture and assembly, inspection and testing at the Contractor's / manufacturer's works, conducting performance and acceptance tests for efficient and trouble free operation, seaworthy packing, supervision of erection, testing and commissioning of LV Switchgears as specified here for the 300 MW CAP DES BICHES Combined Cycle Power Plant at Dakar, Senegal.
- 103.2 It is not the intent of this specification to specify all technical requirements or to set forth requirements adequately covered by applicable codes and standards. The Supplier shall furnish high quality equipment meeting the requirements of this specification and industry standards.
- 103.3 All intrinsic equipment necessary for the proper and safe operation of the equipment and auxiliaries furnished under this scope shall be provided by Supplier.
- 103.4 All data, specification, drawings and documents shall be written in English & French. All units of measurements and dimensions shall be in SI (System International) units, except those used internationally in the English measurement system.
- 103.5 In all the cases, the LV Switchgears shall be designed and selected to ensure high reliability and availability of the complete plant with low forced outage rates and minimal maintenance over the lifetime of the Power Plant, which is defined as NOT less than 25 years.
- 103.6 The following provisions must be noted by the Tenderers.
- a. The Supplier shall bear all costs associated with the conduct of their assessments, studies, visits, and preparation and submission of their Proposals, and the Suppliers' participation in the tender process, without limitation, all costs and expenses related to the Supplier involvement.
- 103.7 Employer and Purchaser shall not be responsible or liable to pay any costs incurred by the Supplier, regardless of the conduct or outcome of the tender process.
- 103.8 Deviations to the Bid if any, whether major or minor should be furnished in the 'List of Declared Deviations' only. In the absence of the deviations listed in the 'List of Declared Deviations', the offer shall be deemed to be in full conformity with the specification notwithstanding anything else stated elsewhere in the offer, data sheet, etc. The hidden deviations or stated / implied deviations in the offer shall not be acceptable OR binding on the Purchaser.
- 103.9 All materials & equipment supplied under this contract shall be new and

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unused

104. DESCRIPTION AND OPERATING CONDITIONS

104.1 The general function of the LV Switchgears are to receive the power from Service Transformers and distribute it to all the drives and equipments in Plant auxiliary system, incase of outage of one Service Transformer, the entire LV Switchgear shall be fed by another Service Transformer through bus Section Breaker. Selection / Sizing of all the equipments / components shall be in order to meet the requirements as specified in various sections of this specification.

104.2 LV Switchgears shall be capable to operate at start-up, shut-down & continuously at all site conditions.

105. SCOPE OF SUPPLY AND EXCLUSIONS

105.1 Scope of supply and exclusions are outlined in Design Requirements (Division 2, Section 200) and Scope Matrix (Attachment 1).

106. SCOPE OF SERVICES

106.1 Refer to Design Requirements (Division 2, Section 200) and Scope Matrix (Attachment 1)

107. TERMINAL POINTS

107.1 Refer to Scope Matrix (Attachment 1)

108. SPECIFIC REQUIREMENTS

108.1 General

- a. Refer to Division 1 sections for requirements regarding Quality Control (Section 104), Product Receiving, Storage and Handling Requirements (Section 106), and Reference Documents (Codes and Standards) (Section 103).
- b. Other items/accessories/equipment not specifically mentioned herein but required for completeness of the system and to meet the intent and requirements of this specification shall also be provided by the bidder.

108.2 Design Aspects

- a. The LV Swtichgears and accessories shall be designed taking into account site conditions and other relevant project data indicated in Site Design Data (Section: 101).
- b. LV Swtichgears shall be capable of delivering the rated output without exceeding the specified temperature limits, under the ambient conditions as specified in Site Design Data (Section: 101).
- c. Auxiliary voltage levels shall be as follows:

- LV auxiliary supply : 400V ($\pm 10\%$), 3 phase, 4 wire, solidly earthed
BSDG : 3 phase, 3 wire 
- DC supply : 125V (+10% to -15%), unearthed 
- UPS supply : 230V, 1 phase , unearthed 
- Power frequency : 50Hz
- Permissible frequency variation : +5% to - 5%
- Combined voltage & frequency variation: 10% (absolute)

d. The LV Switchgears shall be designed considering the following design parameters:

Description	
Type	Indoor, Double front, Multi tier, Metal clad, Fixed type, Air insulated
Ingress protection	IP 42
Busbar	Copper
Rated Voltage (normal / max)	400V / 1000V
Form of Separation	
PCC/MCC Switchgear	Form-3b
PDB/Distribution Boards	Form-2b
Color	RAL 7032
Type of Outgoing CB: Air/ MCCB/ MPCB	Refer attached 400V SLDs
Phase / Wire / Frequency	3 / 4 wire+PE / 50 Hz
Short circuit withstand current rating / duration	50kA (rms) for 1 sec.
Peak withstand current rating	105kA (peak)
One minute power frequency withstand voltage	3kV (rms)
Type of Incomer Circuit breaker	Refer attached 400V SLDs
Type of Protection Relays	Microprocessor based multifunction Numerical relays with MODBUS TCP/IP protocol
Type of Meters	Multifunction, Digital type with MODBUS communication,

Description	
	connction through MODBUS over TCP/IP protocol
Feeder remote control	Controlled from plant PCS; through hardwiring; Interposing Relay shall be located in the Switchgear
Control supply	125V DC
Auxiliary supply	230V AC, 1 phase
Feeder terminal connection	
Incomer	Single core/Multi core, XLPE insulated, Un-Armoured, copper cable with PE conductor or NonSegrigated Phase Bus-Duct
Outgoing	Multi/Single core, XLPE insulated, Un-Armoured, copper cable with PE conductor

108.3 Electrical Design Requirements

- a. Electrical requirements are outlined in the Attachment 1 – Scope Matrix, Attachment 2 – Technical Datasheet.

DIVISION 1 – SECTION 101
SITE DESIGN DATA

PART 1 – GENERAL

109. SITE DESIGN DATA

109.1 Climatic Data

- a. Ambient temperature : Design: 26° C, Max: 41° C, Min: 10 °C
- b. Relative humidity : Design: 70 %, Max: 98%, Min: 32%

109.2 Elevation

- a. Altitude : 20 m + NGA approximately.

109.3 Temperature

- a. Outdoor: Systems and system component design criteria which require outdoor ambient temperature extremes shall use the range from 10°C to 41°C for dry bulb temperatures.
- b. Indoor: Systems and system component design criteria which require indoor ambient temperature extremes shall use the range from 16°C to 46°C for dry-bulb temperatures.

109.4 Wind

All exposed structures or parts of structures shall be designed to withstand wind pressures in all directions, in accordance with Eurocode I.

- Wind direction : North and North – West
- The basic wind speed : 161 km/hr
- Annual Rainfall : 2500mm
- Ground Snow Load : 0 kg/cm²

109.5 Seismicity

For seismicity, the following value shall be considered:

The Dakar region is classified as ZONE 0 in accordance with UBC 97 Soil Profile type SD and Occupancy Category I

**DIVISION 1 – SECTION 102
SHOP DRAWINGS AND PRODUCT DATA**

PART 1 – GENERAL

110. DRAWINGS AND DATA

110.1 General Requirements

- a. The required engineering and design shall begin immediately after Contract award, regardless of the shipping date and shall proceed in an expeditious and orderly manner until complete.
- b. Only project-specific drawings and data shall be submitted to the Purchaser, as described herein. Drawings shall be identified as appropriate.
- c. All drawings and data shall be submitted in metric units.
- d. All documents produced specifically for the project shall be in Microsoft Office format i.e. Word, Excel and Access. Drawings, with the exception of manufacturing and proprietary drawings shall be produced using AutoCAD 2007 or later versions. It's a strong requirement that all the information has to be sent in editable format.

110.2 Document Numbering

All documents submitted shall be identified using KKS tagging.

110.3 Submittals

- a. Documentation and submission procedures will be provided to Vendor after contract award.

110.4 Master Document List

- a. After notification of award, Supplier shall provide the Purchaser with a complete document list. This list shall be updated and resubmitted on a monthly basis for the duration of the work and/or until all drawings have been completed, and shall contain the following information in a format similar to Table 1-1:

**TABLE 1-1
SAMPLE DOCUMENT LIST**

Doc. No.	Document	First Review Submittal (Scheduled)	First Review Submittal (Actual)	Release for Fabrication (Scheduled)	Release for Fabrication (Actual)	Percent Complete
1						
2						

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110.5 Review Procedure

- a. Documentation review procedures will be provided to Vendor after contract award.

111. DOCUMENT SUBMITTAL SCHEDULE

111.1 Refer to Attachment 3, Required Documentation, for a list of the submittals required with the proposal for this specification in addition to the documentation required after award.

112. FINAL RECORD DRAWINGS, DOCUMENTS AND DATA

112.1 At the completion of the project, Supplier shall submit a complete set of all final Supplier and Sub-supplier drawings, including any changes through performance testing.

112.2 For lists or data sheets, Adobe Acrobat PDF format is preferred. Drawing files shall be submitted electronically on compact disk (CD) or Digital Versatile Disc (DVD) with the project name, station name, station unit number, drawing numbers and revision numbers identified on the disk labels together with one full size print of each drawing.

- a. Besides the electronic issue, the number of hardcopies shall be provided as follows:
- b. Three (3) sets of hardcopies plus Three (3) sets of CDs for Operating and Maintenance manuals in English & French
- c. Three (3) sets of hardcopies plus Three (3) sets of CDs for As-Built drawings in English & French

112.3 Software

- a. Any software or hardware upgrades shall be compatible with previous versions for a minimum of 15 years.

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DIVISION 1 – SECTION 103

REFERENCE DOCUMENTS

PART 1 – GENERAL

113. EXTENT

113.1 This Section includes:

- a. Schedule of References.
- b. Quality Assurance Provisions.

113.2 Standards, specifications, manuals, codes and other publications of nationally recognized organizations and associations are referenced herein. References to these documents are to the latest issue date of each document, unless otherwise indicated in this specification or in the Adopted Building Code, together with the latest additions, addenda, amendments, supplements, etc., thereto, in effect as of the date of Contract for the Work.

113.3 Methods, equipment and materials specified herein shall comply with the specified and applicable portions of the referenced documents, in addition to federal, state or local codes having jurisdiction.

113.4 For international projects, the internationally accepted Codes & Standards not listed in this specification shall be acceptable if they are equal to or better than the Codes & Standards referenced herein and/or proposed by the Supplier. The acceptance of those international Codes & Standards shall be determined by the Purchaser and the Purchaser's Engineer.

113.5 Acronyms will be used throughout the text of this specification to refer to the applicable organizations.

114. SCHEDULE OF REFERENCES

114.1 AA – The Aluminum Association, Inc.

114.2 AABC – Associated Air Balance Council

114.3 AAMA – American Architectural Manufacturers Association

114.4 AASHTO – American Association of State Highway and Transportation Officials

114.5 ABMA – American Bearing Manufacturers Association

114.6 ACI – American Concrete Institute

114.7 ADA – Americans with Disabilities Act of 1990

114.8 ADC – Air Diffusion Council

114.9	AGA – American Gas Association
114.10	AGMA – American Gear Manufacturers Association
114.11	AISC – American Institute of Steel Construction
114.12	AISE – Association of Iron and Steel Engineers
114.13	AISI – American Iron and Steel Institute
114.14	ANSI – American National Standards Institute
114.15	API – American Petroleum Institute
114.16	ARI – Air-Conditioning and Refrigeration Institute
114.17	ASCE – American Society of Civil Engineers
114.18	ASHRAE – American Society of Heating, Refrigerating and Air-Conditioning Engineers
114.19	ASME – American Society of Mechanical Engineers
114.20	ASNT – American Society for Nondestructive Testing
114.21	ASSE – American Society of Safety Engineers
114.22	ASTM – ASTM International
114.23	AWS – American Welding Society
114.24	BSI – British Standard Institute
114.25	CE – Electromagnetic Compatibility
114.26	CFR – Code of Federal Regulations
114.27	CGA – Compressed Gas Association
114.28	CPSC – Consumer Product Safety Commission
114.29	CRSI – Concrete Reinforcing Steel Institute
114.30	CSA – Canadian Standards Association
114.31	CTI – Cooling Technology Institute
114.32	EEC – Council of European Communitites
114.33	EEI – Edison Electric Institute
114.34	EIA – Electronic Industries Association
114.35	EJMA – Expansion Joint Manufacturers Association
114.36	EN – European Standards
114.37	EPA – Environmental Protection Agency

114.38	EPRI – Electric Power Research Institute
114.39	EUROCODE – Civil Design Specifications
114.40	FED SPEC – Federal Specifications and Standards
114.41	FM – Factory Mutual
114.42	HEI – Heat Exchange Institute
114.43	HI – Hydraulic Institute
114.44	HMI – Hoist Manufacturers Institute
114.45	IBC – International Building Code
114.46	ICC – International Code Council
114.47	ICEA – Insulated Cable Engineers Association
114.48	IEC – International Electrotechnical Commission
114.49	IEEE – Institute of Electrical and Electronics Engineers
114.50	IFC – International Fire Code
114.51	IGCI – Industrial Gas Cleaning Institute
114.52	IGE – International Institution of Gas Engineers
114.53	IPCEA – Insulated Power Cable Engineers Association
114.54	ISA – International Society of Automation
114.55	ISEA – International Safety Equipment Association
114.56	ISO – International Organization for Standardization
114.57	JIS – Japanese Industrial Standards
114.58	LPC – Lightning Protection Code
114.59	MICA – Midwest Insulation Contractors Association
114.60	MIL – Military Specifications and Standards
114.61	MSS – Manufacturers Standardization Society of the Valves and Fittings Industry
114.62	NACE – NACE International, The Corrosion Society
114.63	NCMA – National Concrete Masonry Association
114.64	NEBB – National Environmental Balancing Bureau
114.65	NEC – National Electrical Code
114.66	NEMA – National Electrical Manufacturers Association

- 114.67 NERC – North American Electric Reliability Corporation
- 114.68 NESC – National Electrical Safety Code
- 114.69 NFPA – National Fire Protection Association
- 114.70 NIA – National Insulation Association
- 114.71 NIST – National Institute of Standards and Technology
- 114.72 OSHA – Occupational Safety & Health Administration
- 114.73 PCA – Portland Cement Association
- 114.74 RCRA – Resource Conservation and Recovery Act
- 114.75 RCSC – Research Council on Structural Connections
- 114.76 RMA – Rubber Manufacturers Association
- 114.77 SAE – SAE International
- 114.78 SAMA – Scientific Apparatus Makers Association
- 114.79 SMACNA – Sheet Metal and Air Conditioning Contractors’ National Association
- 114.80 SSPC – The Society for Protective Coatings
- 114.81 TEMA – Tubular Exchanger Manufacturers Association
- 114.82 TIA – Telecommunications Industry Association
- 114.83 TIMA – Thermal Insulation Manufacturers Association
- 114.84 UBC – Uniform Building Code
- 114.85 UL – UL LLC
- 114.86 USBR – United States Bureau of Reclamation
- 114.87 WB – The World Bank
- 114.88 WHO – World Health Organization
- 114.89 WRC – Welding Research Council Bulletins
115. QUALITY ASSURANCE PROVISIONS
- 115.1 Conform to reference standard by date of issue current on date for receiving bids.
- 115.2 For products or workmanship specified by association, trade or federal standards, comply with requirements of the standard, except when more rigid requirements are specified or are required by applicable codes.

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- 115.3 Obtain copies of standards when required by Contract Documents.
- 115.4 Maintain copy at job site during submittals, planning and progress of the specific work, until substantial completion.
- 115.5 Should a specified reference standards conflict with Contract Documents, request clarification from Consulting Engineers before proceeding.
- 115.6 The contractual relationship of the parties to the Contract shall not be altered from the Contract Documents by mention or inference otherwise in any reference document.

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DIVISION 1 – SECTION 104

QUALITY CONTROL

PART 1 – GENERAL

116. EXTENT

- 116.1 This Section covers the administrative and procedural requirements for quality assurance and quality control.
- 116.2 Testing and inspecting services are required to verify compliance with design and construction specifications and for production of standard products. These services do not relieve Supplier of responsibility for compliance with the Contract Documents.
- 116.3 The Supplier shall have an inspection, testing and documentation program in effect. The Supplier shall be responsible for assuring that its sub-supplier's Quality Control Program meets the specified requirements imposed on the Supplier by the Purchaser.
- 116.4 All manufacturing, processing, testing and inspection operations affecting the equipment or material may be subject to surveillance by the Purchaser or the Purchaser's Representative.
- 116.5 Requirements for Supplier to provide quality-assurance and -control services may be supplemented by provisions of other sections of the specifications.
- 116.6 Other Sections may contain QA activities specific to those work results.
- 116.7 The Supplier is responsible to assure that any activities sub-contracted to lower-tier constructors or agencies are contractually "passed-down" and the Supplier shall assure that their sub-tier entities follow these requirements.

117. REFERENCE DOCUMENTS

- 117.1 Standards, specifications, manuals, codes and other publications of nationally recognized organizations and associations are referenced herein. Methods, equipment and materials specified herein shall comply with the specified and applicable portions of the referenced documents indicated in Section 103, Reference Documents, in addition to local codes having jurisdiction. References to these documents are to the latest issue date of each document, unless otherwise indicated, together with the applicable additions, addenda, amendments, supplements, thereto, in effect as of the date indicated in Section 103, Reference Documents.

118. SUBMITTALS

- 118.1 Submit the following documents for review in accordance with the submittal requirements of Section 102, Shop Drawings and Product Data, and as follows:
- a. The Quality Assurance and Quality Control (QA/QC) Plan.

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- b. Within two (2) weeks prior to beginning of fabrication or as defined elsewhere under Drawings/Data submittal requirements, the following applicable procedures shall be submitted for review and acceptance as applicable:
 - b1. Welding Procedure Specifications (WPS), Procedure Qualification Records (PQR), and welder certifications.
 - b2. Inspection and test procedures conducted in accordance with applicable industry standards such as, but not limited to, ASCE, ANSI, and ASTM.
 - b3. Cleaning, painting, storage, packaging and shipping procedures. Supplier shall attach the Material Safety Data Sheets (MSDS) with all painting/coating procedures.
 - b4. Nondestructive examination procedures and personnel qualifications.
 - b5. The Supplier shall develop and submit an installation plan describing the installation sequence and inspections and/or tests to be performed, for use in determining inspection points that the Purchaser may desire to witness. The Purchaser will advise the Supplier, prior to the start of fabrication, manufacturing, and/or construction of those steps in the fabrication, manufacturing and/or installation process that the Purchaser desires to witness and Purchaser will also identify which of these points that require work to be stopped pending written authorization to proceed from the Purchaser.
- c. Quality documents, such as but not limited to the ones listed below, shall be submitted as required by the Purchaser for review and acceptance:
 - c1. Material certifications.
 - c2. Inspection check sheets and nondestructive examination records.
 - c3. Test records.
 - c4. Manufacturer's data reports, material safety data sheets, etc.
 - c5. Copies of all deviation reports/non-conformance reports and dispositioning records.
 - c6. Certificate of conformance.
- d. Non-conformances and deviations shall be documented in detail and promptly submitted to the Engineer for review, acceptance, and resolution.
- 118.2 The Supplier shall furnish the following documents for the Purchaser's records upon completion of the work:
 - a. Certificate of Compliance, stating that all equipment and materials furnished comply with this Specification and accepted deviations.
 - b. Documents identifying deviations and their acceptance.
 - c. Others as indicated in individual Specification sections.

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119. GENERAL QUALITY CONTROL AND QUALITY ASSURANCE PROVISIONS

- 119.1 Supplier shall ensure that all materials, equipment, products and work are manufactured in accordance with the QA/QC program. The Supplier shall be responsible for ensuring that its sub-supplier's QA and QC programs meet the specified requirements imposed on the Supplier by Purchaser. Supplier shall resolve any comments to the Purchaser's satisfaction, post-award.
- 119.2 The Supplier shall be responsible for ensuring that its sub-supplier's QA and QC programs meet the specified requirements imposed on the Supplier by Purchaser.
- 119.3 Supplier shall develop an inspection point program. The program shall include the manufacturing, inspection and test operations that the Supplier believes may be of interest to Purchaser in demonstrating product quality, whether performed in its or its sub-suppliers' facilities.
- 119.4 Purchaser and/or its designated representative shall have full access to Supplier's and its sub-supplier's facilities for reviewing conformance to the approved QC program and records, and for witnessing of inspections and tests. Purchaser and Engineer shall be notified ten (10) working days prior to the start of the witness points as specified by Purchaser.
- 119.5 The Work shall be designed, fabricated, manufactured and/or performed to meet the service, performance, and minimum level of quality requirements specified herein. The Supplier shall be solely responsible for advising the Purchaser, in writing, of any conflicts between the Project Specification and the Supplier's installation plans, including performance and levels of quality.
- 119.6 Material, fabrication and placing procedures may be subject to inspections and tests conducted by the Purchaser or its designated representative. Such inspections and tests will not relieve the Supplier of responsibility for providing material and placing procedures in compliance with specified requirements. The Purchaser reserves the right, at any time before final acceptance, to reject material, equipment, and workmanship not complying with the specified requirements.
- 119.7 Non-conformances and deviations shall be documented in detail and promptly submitted to the Purchaser for review, acceptance and resolution.
- 119.8 Maintain quality control over supervision, sub-suppliers, suppliers, manufacturers, products, services, workmanship and conditions to produce work in accordance with the Contract Documents.
- 119.9 Examine the areas and conditions under which work is to be installed and the Purchaser in writing of conditions detrimental to the proper and timely completion of the work.
- 119.10 Errors or flaws in the work discerned during construction and which prevent the proper installation shall be corrected promptly. The Supplier shall make immediate substitution of the non-complying material or shall make field

changes to make the non-complying material acceptable.

119.11 For installation steps and tests that the Purchaser desires to witness, the Supplier shall notify the Purchaser ten (10) working days prior to witness points and ten (10) working days prior to hold points as a first notice. A second notice shall be given 72 hours prior to both witness and hold points. Notification of hold or witness points shall not be considered to have been given until the Purchaser has acknowledged receipt of such notification.

119.12 All deviations from this Specification shall be documented and referred to the Purchaser for written disposition.

120. WORKMANSHIP

120.1 Comply with industry standards of the region except when more restrictive tolerances or specified requirements indicate more rigid standards or more precise workmanship.

120.2 Provide suitably qualified personnel to produce work of specified quality.

121. MANUFACTURER'S INSTRUCTIONS

121.1 Comply with equipment supplier's and fabrication Supplier's instructions in full detail, including each step in sequence.

121.2 Should instructions conflict with the Contract Documents, request clarification from the Purchaser before proceeding.

122. COMPLIANCE WITH REFERENCE DOCUMENTS

122.1 For products and workmanship governed by association, trade or federal standards, comply with requirements of this referenced document. General reference documents are listed in Section 103, Reference Documents, of this Specification.

122.2 Have copies of referenced standards on hand during performance of the Work.

123. QUALITY PLAN

123.1 Quality Assurance Program:

- a. The Supplier shall have a QA Program in effect at all times to verify that all items and services, including subcontracted items and services, comply with the requirements of the Purchase Order and its attachments.
- b. This Program shall clearly establish the authority and responsibility of those responsible for the QA Program. Persons performing quality functions shall have sufficient and well-defined responsibility and authority to enforce quality requirements; to identify, initiate, recommend, and provide solutions to quality problems; and to verify the effectiveness of the solutions.
- c. When requested, the Supplier shall submit a controlled copy of its QA or Quality Systems Manual for review and acceptance by the Purchaser. In

addition, when requested by the Purchaser, the Supplier shall submit copies of any associated implementing procedures and/or instructions.

- d. If subsequent to its acceptance, the QA Program is found to be ineffective or inadequate in providing acceptable quality control, the Purchaser reserves the right to require the necessary revisions.
- e. The Quality Program shall describe as a minimum, the following activities as applicable to the Supplier's scope of supply:
 - e1. Document Control System.
 - e2. Procurement Control System.
 - e3. Special Process Control (welding, NDT, etc.).
 - e4. Identification of Materials and Parts.
 - e5. Calibration of Measuring and Test Equipment.
 - e6. Control of Nonconforming Material.
 - e7. Quality Records.
- f. Exceptions to the above Quality program requirements can be made on a case-by-case basis with the Purchaser concurrence.
- g. As applicable to the Supplier's scope of supply, the following information shall be submitted to the Purchaser for review and concurrence, prior to the start of manufacturing:
 - g1. Description/Criteria of acceptance inspections and tests to be performed.
 - g4. Performance test procedures with acceptance criteria.
 - g5. Types of records and reports to be submitted with the finished product.
- h. Once accepted, the Supplier will be monitored during construction for adherence to its Quality Assurance Program.

123.2 Following are examples of the supplier quality monitoring process, and may be used in whole or in part, as determined necessary, to assure that the supplied product meets specification requirements:

- a. Quality Audits.
- b. Review of vendors' "Quality Management Program" (QA/QC Manuals, Fabricator's Inspection Point Program, Shop Inspection/Test Procedures, Special Process/Fabrication Procedures, etc.)
- c. Source surveillance performed during manufacture/fabrication.

124. INSPECTIONS AND TESTS

124.1 General:

- a. The Inspection and Test Plan, provided by the Supplier shall indicate all inspections and tests to be performed with the specified equipment. The Inspection and Test Plan shall refer to the applicable inspection and test instruction, whether for source, incoming, intermediary, final or site inspections and tests. The Supplier's Inspection and Test requirements shall fulfill the minimum inspection and test requirements. The Purchaser, the licensing authorities, as well as their agents shall have the right to witness all tests and inspections carried out in compliance with the Supplier's Inspection and Test Plan. They shall be permitted access the Supplier's shop, where and while all tests or inspections are being performed.
 - b. Shop Tests and Inspections: The shop tests and inspections of the specified equipment shall be carried out according to the Supplier's Inspection and Test Plan, which has been approved by the Purchaser.
 - c. Site Tests and Inspections: The site tests and inspections of the specified equipment shall be carried out according to the Supplier's Site Inspection and Test Plan, which has been approved by the Purchaser.
- 124.2 Material or equipment shall not be shipped before the Supplier presents a Certificate of Compliance, signed and dated by the Supplier's Quality Assurance/Quality Control department. A copy of this document shall be securely attached to the equipment to be shipped. It is important that the equipment arrives at the site with a copy of the Certificate of Compliance attached.
- 124.3 The Supplier's Responsibilities for Sub-suppliers:
- a. The Supplier shall identify in purchase documents to its Suppliers, all applicable quality and QA requirements imposed by this specification on the Supplier and shall ensure compliance thereto. The Supplier has the prime responsibility for sub-supplier surveillance and evaluating and monitoring the implementation of the Quality Assurance Programs of its sub-supplier.
 - b. The Supplier shall submit a copy of its sub-supplier Surveillance plans to the Purchaser for review prior to implementation. Upon request, the Supplier shall submit to the Purchaser copies of the reports of the sub-supplier surveillance activities that it performs on its Suppliers.
- 124.4 Witness and Hold Points:
- a. The Purchaser shall have the right to establish witness and hold points. In addition, the Purchaser may establish witness points to ensure resolution of quality problems.
 - b. Witness points are defined as predetermined points during installation when installation procedures or activities may be inspected or witnessed for compliance with the Contract, specification, and quality requirements.
 - c. Hold points are defined as predetermined points during the installation process when procedures must be inspected and accepted by the Purchaser prior to continuing installation. The Purchaser shall have the right to access

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theSupplier's and its sub-supplier's facilities at any time to verify the progress and quality of the work.

DIVISION 1 – SECTION 105

NAMEPLATES AND TAGS

PART 1 – GENERAL

125. EXTENT

125.1 Section prescribes the minimum requirements for equipment nameplates and tags.

126. GENERAL

126.1 Number equipment and components using KKS numbering system.

126.2 Purchaser's identification numbering system is provided to consistently number equipment and components throughout the generating plant. KKS tagging details for the Equipment offered will be confirmed during detailed engineering.

126.3 Identification numbers shall be consistent with the numbering on drawings, lists, indexes, data sheets, manuals and electronic files.

127. REQUIREMENTS

127.1 Nameplates and tags shall be provided for all equipment, electrical devices, control and instrumentation equipment, and other items of similar nature supplied with a unique identification number and in accordance with the following guidelines.

127.2 The nameplates and tags shall be manufactured of stainless steel or laminated phenolic material with black lettering on white background. Supplier shall engrave the identification information on the nameplates or tags and permanently affix them to the device.

127.3 Nameplates or tags shall be stamped, as a minimum, with information required by applicable code(s) and shall include Manufacturer's name, service conditions, serial number, shop order number, Purchaser's order number, project identification number, designated name of equipment and tag number.

127.4 The nameplate information for Main Equipment shall be in English and French. The nameplate shall be completely visible after installation of equipment.

127.5 Nameplates shall be permanently attached to the equipment using either stainless steel rivets or machine screws.

127.6 All electrical breakers and switches shall have embossed phenolic/plastic labels with a unique breaker/switch number, which shall be taken from the Supplier's drawings, and the name of the component controlled. All separate components and accessories shall be fully identified. Equipment parts and components that require field assembly shall be match marked when necessary to ensure proper erection.

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- 127.7 All separate components and accessories shall be fully identified. Equipment parts and components that require field assembly shall be match marked to ensure proper erection. Part numbers in accordance with the drawings, drawing number and any needed information shall be adequately attached or marked on the part or component.
- 127.8 Vendor shall install stamped stainless steel tags for the following equipment. The tags shall be secured with stainless steel wire with ends crimped together. If the equipment is insulated, tags shall protrude from any insulation. The tag number shall correspond with the identification number given on the Piping and Instrumentation Diagram.
- a. Each electrical panel shall have a nameplate permanently affixed to the cabinet/compartment front with Purchaser's breaker, switch, etc., number.
 - b. Each local instrument cabinet shall have a nameplate permanently affixed to the cabinet front with Purchaser's instrument and cabinet number.
 - c. Instruments with Purchaser's instrument tag number.
 - d. Each subcomponent, such as motors, shall have a tag or nameplate affixed to the item with Purchaser's identification number and equipment name.
 - e. Spare parts and special tools shall be each adequately identified as described in the applicable section of this specification.

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DIVISION 1 – SECTION 106

PRODUCT RECEIVING, STORAGE AND HANDLING REQUIREMENTS

PART 1 – GENERAL

128. EXTENT

128.1 This Section covers the basic requirements for the receipt, handling, and storage of material, products, and equipment received at the Project site based on Purchaser’s requirements. It addresses the following topics:

- a. Receiving, unloading, storage, and protection.
- b. Inspection, verification, and reporting.

128.2 The Supplier is responsible to assure that any activities sub-contracted to lower tier Suppliers or agencies are contractually passed down and Supplier shall assure that their sub-tier entities follow these requirements.

129. RECEIVING, UNLOADING, STORAGE AND PROTECTION

129.1 Purchaser will designate warehouse, storage or laydown areas for material, product, and equipment deliveries.

129.2 The Purchaser shall receive, unload, store, manage, and protect all material, products and equipment delivered based on Purchaser’s requirements. The Supplier shall:

- a. Authorize the delivery vehicle onto the site and escort the delivery while on site.
- b. Direct the safe and proper unloading and handling of the material, products and equipment.
- c. Receive (count, verify, and inspect) the material, products and equipment.
- d. Label, mark and identify the material, product and equipment with visible signage, labels, bar- coding or radio frequency tracking devices.
- e. Properly place the material, products and equipment into approved storage.
- f. Provide and implement proper procedures and controls to protect and maintain the material, products and equipment while in storage in accordance with the Equipment Supplier’s or the Fabrication Supplier’s requirements and the requirements of the Purchaser.
- g. Document the receipt, inspection, and storage maintenance of the material, products, and equipment.

129.3 The Supplier shall provide the heaters and equipment as necessary to control and maintain the temperature and environment as required by the CMS for material or equipment that is sensitive or vulnerable to environmental damage

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or requires heated storage.

129.4 Equipment required to be covered or wrapped during storage shall be maintained by the Supplier including the responsibility to frequently inspect and replace deficient or missing protective coverings.

129.5 Protective Coatings and Preservatives:

- a. Preservative coatings used on components shall be suitable for the conditions normally expected during shipping, storage and throughout the erection period. Supplier shall monitor and reestablish as necessary any preservatives or coatings that have deteriorated and become ineffective.
- b. Each type of preservative used shall be identified as to quality, life expectancy and type. Toxic and hazardous type preservatives shall not be used. Material Safety Data Sheets (MSDSs) shall be provided for all preservatives brought on to the project site. The MSDSs shall be submitted to the Purchaser and shall cover step-by-step procedures, including federal, state and local governing controls for handling and removal of each type of preservative. The material represented on these MSDSs requires approval by the Purchaser prior to the bringing the material on the project site.

129.6 Material, products and equipment shall be stored and tracked by a material log or database system with input capabilities from bar-coding or radio frequency devices so that material location and status is known from the time the material is received, during the time that it is removed from storage, installed in its final location, and until it is turned over to the project start-up group or the Purchaser.

130. INSPECTION, VERIFICATION, AND REPORTING

130.1 The Purchaser shall inspect and verify that products, equipment and fabricated material delivered to the project site are free from obvious damage or defects, correctly identified, and acceptable for their intended use.

130.2 Products, equipment and material shall be inspected visually prior to off-loading from the delivery truck or vehicle. Visual defects shall be handled in the following manner. Small and fragile parts which are susceptible to damage during transport shall be removed from the unit and packed and protected separately.

- a. Any defects in shipping crates or boxes will be noted on the bill of lading and acknowledged (with driver's signature) by the delivery driver.
- b. Once this damage is noted and before the driver and delivery vehicle are released, the crate or box will be opened and the contents inspected for damage.
- c. The Supplier will be contacted immediately, before the driver leaves the project site, for direction as to whether or not the damage item should be returned.

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- d. If the item is accepted with damage, pictures will be taken of the damage and a report (Nonconformance Report or NCR), including the pictures, shall be sent to the Supplier for disposition of the damage or deficiency.
- 130.3 Once the delivery is accepted and offloaded, the material, product or equipment will be verified (receipt inspection) against the requirements of the shop detail drawings, data sheets, project specification or any other documentation available to assure that the equipment or fabricated material has been correctly supplied, properly identified, and suitable for its intended use. This receipt inspection shall be documented using a material control log or database system.
- 130.4 A nonconformance report (NCR) will be filed with the Vendor, with a copy to the Purchaser, for any equipment or fabricated material that does not meet the shop detail drawings, data sheets or project specification.
- 130.5 The Supplier shall track the NCR to closure. No equipment or fabricated material shall be installed with an outstanding unresolved NCR.
- 130.6 It shall be the Supplier's responsibility to work with the Purchaser to correct any discrepancy in the equipment or fabricated material delivered to the project site.
- 130.7 The receipt inspection shall include the following attributes as a minimum:
- a. A review of the packing list for correctness and traceability of the items received.
 - b. A visual examination to determine if there is any physical damage caused by:
 - b1. Rough handling during shipping.
 - b2. Failure of packaging, damage, protective covers or seals.
 - c. A spot check or random measurement of critical dimensions to verify that dimensions of base plates, mounting holes, weld end preps, flanged connections, configuration and orientation conform to the requirements of the design documents.
 - d. A visual examination to verify that the workmanship, materials, welding, machining, and painting conform to the requirements of the design documents.
 - e. Confirmation that marking and identification conforms to specified requirements as follows:
 - e1. Material marking, numbers, piece and part marking.
 - e2. Nameplates, IEC / ASME code stamping, manufacture's name, serial number, tag number, as required.
 - e3. Equipment tags, equipment number, serial number, item number, P.O No., Vendor ID, as required.

- f. Verify acceptable documentation including:
 - f1. Traceability.
 - f2. Legibility.
 - f3. Completeness to the requirements of the product, material, or equipment code or specification.
- 131. SPECIAL HANDLING OF PASSIVATED CORROSION RESISTANT METALS (STAINLESS STEELS, NI-CR ALLOYS, ALUMINUM, AND TITANIUM)
 - 131.1 Passivated corrosion resistant metal (stainless steels, Ni-Cr alloys, aluminum, and titanium) items shall be handled, stored, secured, fit, formed, cut, transported and lifted to prevent iron contamination. The following practices shall be part of any procedures which are developed to achieve this requirement:
 - a. Shipping – do not steel chain to truck bed.
 - b. Lifting – use nylon (or other non-steel) slings.
 - c. Fitting Equipment – shall be made of the same passivated corrosion resistant metal or contact points shall be lined with the same passivated corrosion resistant metal.
 - d. Grinders – shall be marked so that once they are used on carbon steel they shall not be used on passivated corrosion resistant metal.
 - e. Rolling Brakes and Shears – clean all contacting surfaces with a solvent and put cardboard or other membrane between the equipment and the passivated corrosion resistant metal material.

DIVISION 1 – SECTION 107
SPARE PARTS AND SPECIAL TOOLS

PART 1 – GENERAL

132. INTRODUCTION

- 132.1 The overhaul cycle for the plant equipment is four (4) years. Equipment should be capable of running continuously with only routine maintenance and replacement of wearing parts; and no need for replacement of major parts.
- 132.2 Supplier shall supply all spare parts, consumables and special tools necessary for installation, commissioning operation and maintenance of equipment prior to startup.
- 132.3 Supplier shall provide separate lists of spare parts and/or special tools required for:
- Installation and/or erection
 - Performance testing including test equipment
 - Mandatory Spares as per Attachment-1 Scope Matrix (To be quoted as Optional)
 - Startup
 - Two years of operation (To be quoted as Optional)
 - Capital Spares: Those items that would cause more than 90 days downtime if not stocked. (To be quoted as Optional)
 - Routine maintenance between overhauls
 - The first major overhaul
- 132.4 The lists shall:
- Be submitted electronically in Microsoft Excel format.
 - Be inclusive for and applicable to all equipment components, auxiliaries, accessories and materials being furnished under the Contract.
 - Indicate whether the recommended spare is a stock item or special-order item.
 - Be identified with station name, unit no., specification number and equipment name
 - Include the manufacturer of each part, a description of each part (including industry standard part number if available), the assembly or equipment in which each part will be used and recommended quantities to be stocked.

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- f. Classify the relative criticality of parts based on the manufacturer's experience and shall list the recommended lead-time from procurement required for manufacture and delivery of each part.
- g. Be detailed to identify the function of any tools and the specific item or items for which it applies.
- 132.5 Spare parts lists shall include the following items:

TABLE 1-1
SPARE PARTS LIST DATA

Field Name	Description of Information Required
List Number	Sequence number 1, 2, 3, etc.
Manufacturer's Reference No.	Shop order, job order, etc.
Asset ID	Purchaser's asset code
Major Component Description	Name of major component purchased
Major Component Part No.	Supplier's identification number of major component, if any
Subcomponent Part No.	Supplier's subcomponent identification number, if any
Vendor Comments	Supplier comments, if any
Equipment Specification Number	Purchaser's specification number for major component
Manufacturer's Part Description	Spare part description
Manufacturer's Drawing Number	Outline or equipment drawing(s) showing Bill of Material
Item No. on Manufacturer's Drawing	Outline or equipment drawing bill of material item number
Quantity Required for Repair or Rebuild	Number of each spare part required
Unit of Measure	Self-explanatory
Vendor's Recommended Stocking Level	Self-explanatory
Normal Lead Time	Delivery time from release of Purchaser's spare part purchase order
Replacement Frequency	Supplier's recommended repair or replacement frequency
Supplier Contact Information	Company, Name, Position, Address, Phone, FAX,

	Mobile Phone, Email
Unit Price	Self-explanatory
Total Price	For each set of spare parts
Spare Parts Price Validity Date	Date that spare parts price is valid

132.6 A complete operational unit can be identified as a spare part item if it is more cost effective to stock than the separate parts required to maintain the unit.

132.7 Priced Lists shall:

- a. Have pricing valid for one (1) year after equipment delivery.
- b. Indicate the latest date the quoted unit prices are valid.
- b1. Supplier shall submit price escalation formula for recommended and mandatory spares.
- c. Based upon furnishing and delivering the parts f.o.b. at CAP DES BICHES Combined Cycle Power Station, supplier shall indicate the point of shipment, the amount included for freight charges and the lead-time required for delivering the spare parts after order is placed. Spare parts shipping containers will become property of the Purchaser.

133. REQUIREMENTS

133.1 Spare parts shall be packaged and shipped in containers appropriate for the parts. Separate containers shall be used for the spare parts for each major piece of equipment. Where applicable, containers shall be designed and constructed for return shipment of damaged or worn components for repair.

133.2 Spare parts shall be protected from damage due to moisture and dirt accumulation during an extended storage period by use of special coatings, airtight membranes, bags of desiccant or other means acceptable to the Purchaser.

133.3 A weatherproofed itemized list of the contents, spare parts or special tools, shall be attached to the outside of each container and located inside each container. The list shall contain the following additional information:

Supplier Name
Purchase Order Number Specification Number
Quantity
Weight

133.4 Spare Parts

- a. The Purchaser may order all or any of the spare parts at his own discretion.
- b. All spares supplied shall be strictly interchangeable with the parts which they are intended to replace and shall be of the same quality as the original parts.

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- c. Each spare shall be clearly marked with the description, purpose, vendor name, purchase order, specification number, Purchaser's equipment identification number and plant designation code.
- d. If the Supplier follows the practice of assigning his part numbers to parts purchased, Supplier shall provide a cross-reference between the Supplier's part number and the part number of the original Supplier. All parts must be shown clearly on outline and/or installation drawings.
- d1. For each ten (or less) identical items furnished by Supplier as part of the system design, one spare item shall be furnished and the total quantities furnished shall be identified in the Spare Parts List. A maximum of 3 spare items for each spare item type shall be furnished.
- e. Small items shall be packed in sealed transparent plastic bags with desiccator packs as necessary.
- f. When more than one item is packed in a single case or container, a general description of the contents shall be provided on the outside of the case or container and a detailed list enclosed.
- g. All cases, containers and packages shall be suitably marked and numbered for identification purposes. Each container shall be marked with the large painted legend as follows:

WEST AFRICAN ENERGY

CAP DES BICHES COMBINED CYCLE POWER PLANT PROJECT

SPARE PARTS FOR (EQUIPMENT NAME & EQUIPMENT NUMBER)

133.5 Consumable Parts and Materials

The Supplier shall supply a sufficient quantity of consumable parts and materials such as gaskets, seals, anti-corrosion paint for weld bends, handhole covers, etc. as may be required prior to Commercial Operation.

133.6 Tools and Equipment

- a. Any special tools and test equipment required to adjust, dismantle, re-assemble or maintain any equipment supplied shall be provided.
- b. Tools and test equipment shall be of suitable quality.
- c. The maintenance tools include all special handling rigs, bars, slings and cable.
- d. All special tools required for installation/erection/commissioning shall be shipped with their associated equipment. All special tools for normal maintenance will be turned over to Purchaser no later than Commercial Operation.
- e. Tools and test equipment shall be neatly arranged in a box and shall be shipped to the site in a suitable container clearly marked with the name of the

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equipment for which the tools and test equipment are intended. The tools shall be shipped in a separate container along with the equipment.

- f. Maintenance tools for each piece of equipment shall be boxed separately and the boxes shall be marked with the large painted legend as follows:

WEST AFRICAN ENERGY

CAP DES BICHES COMBINED CYCLE POWER PLANT PROJECT

MAINTENANCE TOOLS FOR (EQUIPMENT NAME & EQUIPMENT NUMBER)

- g. Erection tools shall remain the property of the Supplier and all shipping costs to and from the job site shall be at the Supplier's expense. Erection tools for each piece of equipment shall be boxed separately. Erection tools shall not be boxed with maintenance tools.
- 133.7 Special storage requirements and/or special maintenance requirements for storage shall be specified, described in detail and shipped with the spare parts and/or special tools.
- 133.8 SOURCE OF SUPPLY

Supplier shall provide assurances that their Suppliers shall provide domestic support services (both technical and administrative) and parts supply for all imported equipment, such services and parts to be provided within the time frames typically available for similar equipment used in Senegal.

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DIVISION 1 – SECTION 108

DESIGN FOR ACCESSIBILITY, OPERATION AND MAINTAINABILITY

PART 1 – GENERAL

134. SCOPE OF SUPPLY

- Design of LV Switchgear
- Drawings and documentation in English and French
- Manufacturing and Assembly in Factory
- Factory Testing and Inspection (Routine and Factory Acceptance Test)
- Type Test Certificate for LV Switchgear with same type and rating (Repetition of test is not required)
- Special Tools if required
- Spare parts for installation, test and commissioning
- Installation instructions and O/M Manuals in English and French
- Supervision for Erection (To be quoted as optional with required days)
- Supervision for Test & Commissioning including test tools (To be quoted as optional)

135. EXTENT

135.1 This Section prescribes the requirements for design and/or construction that includes facilities and space to permit acceptable operation and maintenance of the equipment without shutdown or removal of other critical systems or components and adequate operator interface. Ease of access to the various pieces of equipment for operation and maintenance is a high priority.

135.2 Adequate layout area shall be provided to permit laydown of disassembled components.

135.3 Clearances for walkways, equipment removal areas and access areas for maintenance of equipment shall be provided.

135.4 Equipment shall be installed so it is readily accessible for operation and maintenance. Equipment shall not be blocked or concealed.

136. OPERATION AND MAINTENANCE CONSIDERATIONS

136.1 Operability:

- a. Switchgears will be of Indoor, Double front, single/multi tier, metal clad.
- b. 40°C shall be considered as an ambient temperature prevailing the switchgears

while carrying rated current, as Design ambient temperature for sizing the equipment/device/bus rating. Temperature rise shall be as per applicable IEC standards. The switchgear will be sized to carry the full load current of Service Transformers. 10% design margin will be considered in arriving at the current rating. Based on the fault level, the short circuit withstand rating shall be rated for 50 kA for 1 sec. One minute Power Frequency withstand voltage will be 3 kV (rms).

- c. Equipment system design selections shall be based on minimizing the amount of operator attention.
- d. Equipment operation shall be adequately monitored to provide control room operators with all information required for efficient, safe and easy operation of the plant.
- e. Instrumentation that indicates temperatures, etc., or indicating the position or status of equipment shall be clearly visible and readable to operating personnel and installed in accordance with the manufacturer's recommendations.

136.2 Accessibility:

- a. To ensure safety during maintenance work all parts of the main circuit, to which access is required, shall be provided with facilities for connecting removable earthing device, after opening the enclosure.
- b. Clearances to the front and rear of equipment shall be not less than the greater of the distance required by the National Electric Code (NEC) or the manufacturer's recommendation.
- c. A standard mechanical user interface, ergonomically Positioned at a convenient height. It must be visible directly without opening of doors etc.
- d. The user interface comprises all the mechanical, panel- related interfaces and continuous interrogating interlocks.
- e. All the basic mechanical ON/OFF of CB, manual spring charge of CB must be possible without opening the door to ensure the operator safety.
- f. The mechanical control panel is located at an optimum height for operation and arranged in a recessed position on the switchgear front. Thus, the operating area is clearly visible while no control elements protrude from the switchgear front.
- g. The position of the individual elements shall be selected according to their function, i.e. according to their allocation to the corresponding device functions. The elements which form part of a switching device, such as position indicators, crank ports or mechanical push buttons, are visually linked by a specific pattern and integrated in a mimic diagram.

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- h. Mechanical operation is performed the same way as with the habitual operation with stationary switching devices. Separate control elements and mechanical switch position indicators are available for the following functions:

- Circuit-breaker ON - OFF

136.3 Maintainability:

- a. Equipment shall be designed to be maintained in place with minimum disassembly of surrounding equipment.
- b. The basic switchgear unit is to be designed for free - standing installation within the switchgear room.
- c. Equipment arrangements shall be designed for maximum equipment accessibility and to allow the following types of access:
 - c1. Space shall be provided to allow plant personnel easy access to all equipment, which may require maintenance.
 - c2. Space shall be provided to allow unobstructed access for maintenance tools and equipment required for maintenance on permanently installed equipment.
 - c3. Ample space shall be provided to allow removal and laydown of any equipment that cannot be maintained in place or may require replacement.
 - d. Lifting eyes shall be provided on equipment to facilitate installation and removal for maintenance.
 - e. Techniques for minimizing corrosion of structures and equipment exposed to chemically or environmentally corrosive atmospheres shall be incorporated into the equipment design. Removable panels with lifting eyes on enclosures shall be provided where required.
 - f. Where feasible, similar equipment shall be provided by the same Contractor to minimize spare parts inventories and to minimize the number of different Contractors' equipment that plant personnel must be capable of maintaining.
 - g. Equipment and systems shall be provided with isolation capability for personnel safety during maintenance activities. Design shall facilitate lock out, tag out, double block and bleed, etc., to satisfy requirements of OSHA 19 10.269 and 147.
 - h. Equipment removal paths shall be maintained clear of obstructions and interferences.

137. REFERENCE DOCUMENTS

- 137.1 Standards, specifications, manuals, codes and other publications of nationally

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recognized organizations and associations are referenced herein. Methods, equipment and materials specified herein shall comply with the specified and applicable portions of the referenced documents, in addition to federal, state or local codes having jurisdiction. References to these documents are to the latest issue date of each document, unless otherwise indicated, together with the applicable additions, addenda, amendments, supplements, thereto.

- 137.2 OSHA – Occupational Safety and Health Standards:
- a. 29 CFR 1910.147 – The Control of Hazardous Energy
 - b. 29 CFR 1910.269 – Electric Power Generation Transmission and Distribution
- 137.3 For international projects, the internationally accepted Codes & Standards not listed in this specification shall be acceptable if they are equal to or better than the Codes & Standards referenced herein and/or proposed by the Supplier. The acceptance of those international Codes & Standards shall be determined by the Purchaser.

DIVISION 2 – SECTION 200

DESIGN REQUIREMENTS

201. The Supplier shall comply with the technical requirement of the Specification and additional requirements as indicated below:

202. DESIGN CRITERIA

202.1 The equipment to be furnished under this specification shall be in accordance with the applicable section of the latest version of the following IEC standards of latest edition including amendments, except where modified and /or supplemented by this specification.

IEC 60947	LV switchgear and Control Gear
IEC 60158	LV Control Gear
IEC 60408	LV Air-break Switches, Air-break Disconnectors, etc.,
IEC 60269	LV Fuses
IEC 60099-4	Surge arresters - Part 4: Metal-oxide surge arresters without gaps for a.c. systems
IEC 60099-6	Surge arresters - Part 6: Surge arresters containing both series and parallel gapped structures - System voltage of 52 kV and less
IEC 61869	Instrument Transformers – All parts
IEC 60255	Measuring relays and protection equipment
IEC: 60529	Degrees of protection provided by enclosures (IP Code)
IEC 60038	IEC Standard voltages
IEC 60909	Short-circuit currents in three phase a.c. systems
IEC 61439-1	General rules
IEC 61439-2	Power Switchgear and control gear assemblies
IEC 61439-3	Distribution Boards
IEC 61439-5	Assemblies for power distribution
IEC 61439-6	Bus Trunking System
IEC TR 61641	Enclosed low voltage switchgear and control assemblies
	Guide for testing under conditions of arcing due to internal fault

202.2 Design ambient air temperature shall be considered as 40°C for sizing the electrical equipment and relative humidity of 98% maximum.

202.3 Insulation level for 400 V System:

- a) One minute Power Frequency withstand voltage : 2.5kV (rms)

202.4 Short circuit withstand rating

- a) Short circuit rating : 50 kA for 1 second
b) Dynamic withstand rating : 105 kA (peak)

202.5 Internal arc withstand rating shall be not less than 50 kA (rms) for the duration of 0.5 second.

202.6 Neutral Grounding System

- a) Neutral of the 400 V systems will be solidly grounded.

202.7 Current Rating:

- a) Ambient temperature prevailing inside the switchgear while carrying rated current, when the outside air ambient temperature is 40°C shall be considered as Design ambient temperature for sizing the equipment/device/bus rating. Temperature rise shall be as per applicable IEC standards. A design margin of not less than 10% shall be considered for sizing various devices at minimum operating voltage.

MCC/PDB Description	Location- Indoor/Outdoor	Ambient temperature (in deg C)
400V GTG#1 MCC	Indoor	30°C
400V GTG#2 MCC	Indoor	30°C
400V Unit Switchgear	Indoor	30°C
400V STG MCC	Indoor	30°C
400V Station Switchgear	Indoor	30°C
400V DG PCC	Indoor	30°C
400V ACDB	Indoor	30°C
400V MLDB	Indoor	30°C
400V Welding DB	Indoor	30°C
400V Main Plant HVAC PDB	Indoor	30°C
400V Auxiliary Boiler MCC	Indoor	30°C
400V HRSG MCC #1	Indoor	30°C



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400V HRSG MCC #2	Indoor	30°C
400V MLDB	Indoor	30°C
400V RMS Area PDB (RMS & STP Loads)	Indoor	30°C
400V Admin, Control Building PDB	Indoor	30°C
400V Switchyard PDB	Indoor Non-AC	40°C
400V Workshop Area PDB	Indoor	30°C
400V ACC Switchgear #1	Indoor	30°C
400V ACC Switchgear #2	Indoor	30°C
400V Auxiliary MCC	Indoor	30°C
400V WTP MCC	Indoor	30°C
400V MLDB	Indoor	30°C
400V Welding DB	Indoor	30°C
400V WTP Area HVAC PDB	Indoor	30°C

- b) The continuous current rating of the incomers/tie feeders/main bus bars of PCC shall be based on the name plate rating of the upstream transformer with 10% margin at design ambient condition, rounded off to next higher standard rating unless specified otherwise.
- c) For the Switchgears having two bus sections, each bus section & incomer shall be rated for load requirements of both the bus sections for continuous current rating.
- d) MCCB / MPCB contactor and Electronic overload relay for IE2 motor feeders shall meet type-2 coordination as per applicable standard.
- e) Similar equipment and components shall be of same make. Equipment of same type and rating shall be interchangeable.
- f) Bidder to provide switchgears of type tested design for the offered rating.

202.8 Protection

- a) The Vendor shall be responsible for selection of protective devices so as to provide optimum protection and discrimination of various circuits and equipment including the safety of the devices. For motor feeders, coordination of overload relay and MPCB/MCCB with the thermal characteristics of the motor shall be ensured.

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- b) The incomer modules shall be interlocked with their upstream breaker such that they can be closed only when upstream breaker is closed and trip automatically when upstream breaker is tripped.
- c) Each breaker module shall be provided with multifunction numerical relay for protection. Relays shall have built in protection, control & metering and communication modules shall be suitable for MODBUS TCP/IP protocol.
- d) The following protection as a minimum shall be provided for ACB operated feeders.
 - 50 - Short circuit protection.
 - 50N - Instantaneous earth fault
 - 51 - IDMT over current protection
 - 51N - IDMT Earth Fault over current protection
 - 51N1 - IDMT Standby earth fault protection. (Only for transformer incomers)
 - 64R - Restricted Earth fault protection (only for transformer incomers).
 - 25C - Check Synchronising.
 - 27 - Under voltage protection (as applicable)
 - 94 - Electrical anti pumping feature
 - 86 - Hand reset high speed lockout relay
 - 95 - Trip circuit supervision relay (separate relay)
 - 98 - VT supervision / MCCB / MCB fail (as applicable)
 - 99M - Motor comprehensive protection (as applicable)
- e) ACB controlled motor feeders shall be provided with Numerical motor protection relay along with the above protections as applicable
- f) Separate Hand reset lockout relay (86) to be provided for each Breaker. In addition self reset type lockout relay to be provided on Bus PT Panel for Under voltage tripping of the ACB Operated motor feeders.

202.9 Changeover Scheme

- The following change-over features are envisaged for PCC (Unit & Station Switchgears):
 - i. Manual planned transfer (momentary paralleling mode), the transfer is initiated manually to other bus during normal operation of the unit (in case one Service Transformer is to be taken out from service) and transfer takes place through momentary paralleling of two supplies
 - ii. Protective transfer shall be initiated automatically; on operation of master trip relay transfer will be blocked. Transfer will initiate through under voltage relay operation or service transformer unit relay protection relay operation. Transfer will be blocked in case of incomer feeder relay operation.
 - iii. Slow bus transfer shall be applicable for Auto transfer during under voltage by continuously monitoring the system parameters.

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- For MCC & other PDBs, only manual transfer via interlocks shall be provided.
- For ACC Switchgear, Microprocessor based numerical automatic changeover scheme for changeover of supply shall be provided for the incomers. The following change-over features are envisaged.
 - i. Manual planned transfer (momentary paralleling mode), the transfer is initiated manually to other bus during normal operation of the unit (in case one Service Transformer is to be taken out from service) and transfer takes place through momentary paralleling of two supplies
 - ii. Protective transfer shall be initiated automatically, on operation of Master trip relay transfer will be blocked. Transfer will initiate through under voltage relay operation or service transformer unit relay protection relay operation. Transfer will be blocked in case of incomer feeder relay operation.
 - iii. Slow bus transfer shall be applicable for Auto transfer during under voltage, bus by continuously monitoring the system parameters.

202.10 Controls

- a) Incomers and bus sections of PCCs shall be breaker controlled. PCC defined elsewhere in this specification refers to LV switchboard boards connected to load center transformer which may contain outgoing power feeders as well as outgoing motors feeders.
- b) Outgoing feeders of PCC shall be controlled by breakers / moulded case circuit breakers (MCCBs) Motor protection circuit breakers (MPCBs) /contactors depending on their ratings / applications.
- c) Incomers and bus sections (if applicable) of MCCs shall be breaker controlled. Outgoing feeders of MCCs shall be controlled by MCCBs / contactors / applications.
- d) Feeders rated above 400A shall be controlled by air circuit breakers (ACBs), unless specified otherwise.
- e) Feeders rated upto 400A may be controlled by MCCBs. However ACB shall be used for 400A feeders where changeover is envisaged.
- f) ACBs shall be used for motor feeders rated $\geq 75\text{kW}$ & $< 200\text{kW}$ for direct-on-line (DOL) starting.
- g) MCCB Contactor starter units shall be used for motor feeders rated $> 10\text{kW}$ & $< 75\text{kW}$ for DOL starting.

Motors rated $\leq 10\text{kW}$: MPCB Contactor controlled



- h) Feeders of 400V PCC/MCC shall be controlled from PCS or Local Push Button (LPB) Station or Local Control Panel (LCP).

- i) Remote selection shall enable control from PCS/PLC/LCP.
- j) All motors shall be controlled and monitored from the PCS system.
- k) The control/interlock schemes for various types of feeders shall be commensurate with their applications.
- l) Various numerical relays, and digital multi-function meters shall be connected to plant PCS through Ethernet switch located in each switchgear. Intergation with plant PCS from the ethernet switch connection onwards is by others. Necessary communication relevant files will be shared during detailed engg.
- m) Incomer breaker shall be interlocked with their upstream breaker such that they can be closed only when upstream breaker is closed and trip automatically when upstream breaker is tripped.
- n) All the breaker controls from PCS to Switchgear and feedback from Switchgear to PCS shall be hardwired.
- o) All analog signals from Digital Meters shall be converted on MODBUS over TCP/IP protocol and communicated to PCS. All numerical relays shall be communicated to PCS through MODBUS TCP/IP protocol. All Relays & meters in Panel shall be connected to Ethernet Switch (with Two Ports) for further Connection to Purchaser's Automation System.
- p) All hardware/software required for interfacing the with the PCS system shall be supplied by the Supplier.

203. CONSTRUCTIONAL FEATURES

203.1 Construction

- a) PCCs shall have duplicate incomers and bus section. The outgoing feeder panels shall consist of Air circuit breakers / MCCBs / Contactors / MPCBs depending on their ratings / applications.
- b) MCCs shall have duplicate incomers and bus section, unless indicated otherwise. The outgoing feeder panels shall consist of MCCBs / MPCBs / Contactors depending on their ratings / applications.
- c) DB shall be metal enclosed, indoor, floor mounted free standing type.
- d) Switchboard frames, load bearing members, doors and covers shall be fabricated using rolled steel sections and cold rolled steel sheets. The thickness of sheet shall be not less than 2.0mm. Gland plate thickness shall be minimum 3.0mm made out of hot rolled or cold rolled steel sheets for muticore cable termination and for single core cables termination, it shall be aluminium or nonmagnetic material.
- e) PCC / MCC / distribution boards shall be fixed type with compartmentalized construction. However ACB shall be withdrawable type.

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- f) All PCCs / MCCs shall be double front type, unless specified otherwise. However, breaker feeders shall be single front type.
- g) All DB shall be Single/double front type, the same will be confirmed during detailed engg.
- h) The switchboard shall be provided with an integral base channel of at least 50mm height. Switchboard shall be mounted on channels embedded in concrete by fixing with screws base channels.
- i) All switchboards shall be compartmentalized (conforming to Form 3B of IEC 61439) with uniform height not exceeding 2450mm. Module size of switchboards shall not be less than 200mm.
- j) The switchboards shall be arc proof tested. In the event of an internal fault in a functional unit, the damage shall be confined to that unit alone, so that the busbar system and all other functional units remain fit for further service. Switchgear assemblies shall comply with IEC TR 61641 with regard to internal arc testing.
- k) The arrangement of feeders in the switchboard shall take into consideration the number and size of cables required for the feeders. Incomer and bus section ACB shall be limited to one per panel. ACB outgoing feeders shall be limited to two per panel for ratings upto 1600A. For ratings 2000A and above, the number of outgoing feeders shall be limited to one per panel. 
- l) Bus duct entry shall be from top and cable entry shall be from bottom for PCCs/MCCs/DBs. Switchboards shall be divided into distinct vertical sections (panels), each comprising of the following compartments:
 - Main bus bar compartment
 - Switchgear/feeder compartment
 - Cable alley of at least 300mm width
 - Auxiliary bus bar compartment
 - Control compartment for relays for ACB feeder
- m) Switchboard shall be of modular design, consisting of units or modules, segregated into single or multi-tier compartments. Each unit shall be placed in individual compartment. All units grouped together shall form the complete switchboard. Completed switchboard shall be of flush-fronted design having a neat appearance and shall be readily extensible.

Door shall be provided at the front of each compartment to give full access to all the components and the compartment. The switchboard shall consist of dust and vermin-proof cubicles.
- n) The switchboards shall be designed to prevent contact with live parts both within the modules and in the cable alley. Following are the minimum segregations to be provided amongst the different parts of the switchboard:
 - Main horizontal bus bar chamber and vertical bus bar chamber
 - Between different modules

- Between modules and cable chamber
 - Between auxiliary bus bars and the rest of the switchboard
 - Between vertical bus bars and modules
- o) Interlock between compartment door and modules shall be provided such that the door cannot be opened without switching off the power supply to the module. Defeat interlock shall be provided for the units comprising of MCCB/MPCB as a means of isolation, such that it is possible to open the door with device ON and the unit in ‘SERVICE’ position. The following shall apply under these conditions:
- It shall not be possible to close the door till the interlock has been reinstated
 - Attempted withdrawal of the unit under such conditions shall be cautioned against by providing warning label affixed at an appropriate place
- p) Switchboards shall be equipped with thermostatically controlled space heaters in each panel (separate for front and rear in case of double fronted panels having separate cable alleys) so as to maintain interior temperature 5°C above the ambient. They shall have manual disconnect switch and MCB for protection.
- q) All LT switchgear modules shall be provided with Pad lock and Key arrangement.
- r) One equipped feeder shall be provided for each / rating in MCC as spare. 
- s) 20% of control terminals in each module shall be provided as spare subject to a minimum of 2 terminals.

203.2 Feeder Modules

- a) Drawout modules shall have the following features:
- Withdrawal of units shall be possible without disconnecting wires.
 - Drawout units of similar type and rating shall be interchangeable.
 - Secondary connections to the drawout unit shall be through self-aligning and self-isolating contacts. In case of circuit breakers, easily disconnecting type plug and socket shall be acceptable in place of self-aligning and self-isolating type contacts.
 - With the drawout module fully removed from its housing, live parts exposed shall either be finger proof or shall have means to shroud them totally to prevent contact. Suitable add-on type shrouds shall be provided where automatic shutters or in-built shrouds are not provided
- b) All single front switchboards shall be provided with single leaf, hinged or bolted covers at the rear. The bolts shall be of captive type. The covers shall be provided with ‘DANGER’ labels.
- c) Drawout modules shall have distinct ‘SERVICE’ and ‘TEST’ positions. The equipment pertaining to a drawout type module shall be mounted on a fully withdrawable chassis, which can be drawn out without having to unscrew any wire or cable connection. Suitable arrangement with cradle/rollers and guides shall be provided for smooth movement of the chassis.

- d) Each switchboard shall be provided with removable type gland plate, which shall cover the entire cable alley. For all single core cables, gland plate shall be of non-magnetic material.
- e) Minimum air clearance for horizontal and vertical bus bar termination shall be 25mm between phases and 20mm between phase and earth. For all other components the clearance shall be 10mm minimum.
- f) Switchboard shall be of unit construction suitable for splitting into sections for shipping to site and to be correctly re-erected on prepared foundations without skilled supervision. The individual shipping section shall comprise not more than three verticals. For each shipping section adequate lifting facilities such as hooks or lugs for ease of handling on site shall be provided. Removal of such hooks or lugs shall leave no opening in the switchboard.
- g) The switchboard shall be easily extensible on either side by the addition of vertical sections. It shall be possible to extend the switchboard, irrespective of the type of end panel and the design shall be such as to permit addition of extension panels of a type other than the type of end panel. End bus bar fish plates shall be provided and the same shall be shipped as loose items.
- h) Front access shall be available to all components in each cubicle which require adjustment, maintenance or replacement. Checking and removal of components shall be possible without disturbing adjacent equipment. All auxiliary equipment shall be easily accessible. Setting of relays shall be possible without de-energizing other equipment. Access shall be available to all cable glands and multi-core terminal blocks by means of sheet steel hinged doors, designed to give the maximum possible access to the cable terminations.
- i) All interior barriers shall be manufactured from sheet steel. Each compartment shall be constructed and segregated to confine any damage caused by an internal fault to that compartment. Adequate barriers shall permit personnel to work safely within an empty switching device compartment or one from which the switching device assembly has been temporarily removed with bus energized.
- j) The arrangement of feeders in the switchboard shall take into consideration the number and size of cables required for the feeders. Removable type un-drilled gland plates shall be provided on bottom of the panel. Suitable provision shall be made for clamping cables inside the switchboard. The cable terminations inside the cable alley shall be completely shrouded, so that it shall be possible to work on any one of the terminations by switching OFF the corresponding feeder switch only.
- k) The arrangement of the feeders shall ensure that operating handle of the switch/breaker shall be above 350mm but below 1800mm from finished floor level.
- l) Each panel section of PCC/MCC/DB shall be equipped with the following as required:
- Illumination lamp with door switch
 - 13A 3pin socket with miniature circuit breaker (MCB) protection

203.3 Bus Bars and Insulators

- a) Each PCC/MCC/DB shall be provided with three phase bus bars and one neutral bus bar as specified. DC distribution boards shall have two bus bars.
- b) Bus bars (including neutral bus bar) shall be capable of carrying the short time current for the duration as specified, without any damage. Current carrying capacity of neutral bus bar shall be minimum half that of the phase bus bar.
- c) Main bus bars shall be of interleaved type and connections shall be prominently marked and displaced for standard sequence counting from rear to front, top to bottom, or left to right as viewed from the switching device operating mechanism side. For double front switchboards, the 'front' for the purpose of counting the phase sequence of bus bars shall be as per the approved general arrangement drawings.
- d) For double front type of board, separate vertical bus bars for each front shall be used.
- e) Bus bars shall be made of high conductivity copper. Vertical bus bar for all multi tier modules shall be of copper. They shall be supported on insulators of material such as fiber reinforced polyester or epoxy cast resin. The insulation level of neutral bus bar shall be same as that of phase bus bars.
- f) All bus bars shall be adequately supported by non-hygroscopic, non-combustible, track-resistant and high strength insulators. All bus bar joints shall be provided with high tensile steel bolts, Belleville/spring washers and nuts. All bus bars and bus taps shall have HRPVC sleeves, colour coded as follows:

Brown, Black & Grey for U, V & W phases respectively & Blue for neutral
Brown and Blue for Phase & Neutral respectively

- g) Contact surfaces at all joints shall be properly cleaned and anti-oxide grease applied to ensure an efficient and trouble free connection..
- h) Bus bar shall have minimum cross section corresponding to current density of 0.8A/sq.mm at rated current and maximum permissible temperature. All bus bars and their main current carrying connections shall have the same sectional area through out their length. The bus bars shall be colour coded such that on removal of any door the phases shall be identifiable.
- i) Bus bars and connections shall be secured in such a manner that the insulators are not subjected to bending forces under short circuit conditions. When multiple switching devices are combined in tiers for a vertical unit, the droppers shall be able to carry the total current resulting from the combination of all switching devices. The vertical bus bars shall be completely shrouded with the cutout for connection tapping. The tap offs from vertical bus bars to the switch terminals must be able to withstand the board fault level.
- j) Panels which receive bus duct at the incomer shall have the main bus bars extended on the top of the panel of about 150mm to 175mm for enabling connection to bus ducts.

- k) Auxiliary bus bars each of minimum 25sq.mm thick electrolytic tough pitch copper shall be provided for the following applications:
- 230V AC supply for panel space heater/motor space heater/panel illumination
 - 125V DC control supply for breaker tripping and closing, indication circuits
 - 230V AC control supply for contactor operated motor starter control circuits as applicable.

203.4 **Control supply and space heater supply**

- a) PCC and ACB operated MCC shall receive 2 nos. 125V DC feeders for control supply. Suitable diodes shall be provided to derive control supply through diode auctioneering. Each panel shall receive control supply through bus wires tapped off from auxiliary copper bus bar through MCCBs/MCBs provided in the respective panel. It shall be possible to isolate any panel without disturbing the power supply to other panels. Each sub-circuit shall have separate MCB.
- b) An under voltage relay to monitor control supply shall be provided. A contact of the relay shall be wired to the terminal for external use. Control Supply Failed' indication shall be provided.
- c) PCC/MCC shall derive 230V AC control supply through 400/230V transformer. 2 nos. 400V/230V transformers shall be provided with changeover arrangement. Changeover shall be blocked for control bus bar fault. The control transformers shall be of insulation class 'B' or better. The sizing of control transformers shall be carried out by the Vendor considering the actual load of power contactors, auxiliary contactors, indicating lamps and other equipment including remote auxiliary relays and lamps in the circuit with 20% design margin. One pole of secondary winding shall be earthed.
- d) 230V AC Space heater supply shall be derived from 400/230V control supply transformer located one in each bus section. Necessary MCB to isolate and distribute the supply shall be provided. The control transformers shall be of insulation class 'B' or better. The sizing of control transformers shall be carried out by the Vendor considering the actual load of space heaters, panel lighting & sockets. One pole of secondary winding shall be earthed.
- e) For ACB feeders the 125V DC supply shall be extended in test position also for testing of the ACB operation.
- f) All motors rated 50kW and above shall be provided with power supply for space heater. Circuit for motor space heater shall be wired through auxiliary NC contact of breaker/contactator/MCB.
- g) Each panel of PMCC/MCC/DB shall be equipped with the following as required:
- Thermostatically controlled space heater(s).
 - Illumination lamp with door switch
 - 5A 3pin socket (Type-E) with MCB protection.

203.5 **Air Circuit Breakers**

- a) Air circuit breakers shall be stored energy, trip free, air break and horizontal drawout type and shall have fault making and breaking capacities as specified. There shall be 'SERVICE', 'TEST' and 'ISOLATED' positions for the breakers. In 'TEST' position the circuit breaker shall be capable of being tested for operation without energizing the power circuits i.e. the power contacts shall be disconnected, while the control circuits shall remain undisturbed. Locking facilities shall be provided so as to prevent movement of the circuit breaker from the 'SERVICE', 'TEST' or 'ISOLATED' position. It shall be possible to close the door in 'TEST' position.
- b) Air circuit breakers shall be provided with minimum 6 normally open (NO) and 6 normally closed (NC) potential free auxiliary contacts. These contacts shall be in addition to those required for internal mechanism of the breaker and shall be directly operated from breaker operating mechanism. Separate limit switches, each having required number of contacts shall be provided in both 'SERVICE' and 'TEST' positions of the breaker.
- c) Breakers of different ratings shall not be interchangeable. Front fascia cut-outs for all ratings of breakers shall be identical.
- d) The spring charging mechanism of the circuit breaker shall be motor operated. The close/trip control switch shall be interlocked to trip before close. The closing and tripping circuits shall be self-opening on completion of their respective functions irrespective of the position of the control switch. Manual closing devices shall also be provided. The circuit breaker shall be electrically and mechanically trip free. For all circuit breakers electrical anti-pumping device shall be provided.
- e) The main and secondary isolating contacts of the circuit breaker shall be of self-aligning type. The main isolating contact shall have continuous rating equal to the rating of the breaker. The secondary isolating contact shall be of wiping contact type.
- f) The fixed portion of the circuit breaker shall have rail arrangement over which the chassis can move smoothly. Proper mechanical indication shall be provided for the three positions ('SERVICE', 'TEST' and 'ISOLATED') without opening the compartment door. It shall be possible to bring the circuit breaker to isolated position with the help of external lever without opening the compartment door. A stop block shall be provided on the slide rails to prevent the movement of the circuit breaker out of the compartment when it reaches the isolated position so that any accidental fall can be avoided. No control and metering equipment shall be mounted on breaker compartment door (i.e. access to control and metering equipment shall be available at all time without opening breaker compartment door). The circuit breaker can be padlocked in OFF position with door interlock defeat facility.
- g) The following interlocks shall be provided on the circuit breaker:
 - It shall not be possible to withdraw the circuit breaker from the service position with the contacts of the breaker closed
 - It shall not be possible to close the circuit breaker unless any one of the three

- positions is located
- It shall not be possible to open the compartment door when the circuit breaker is ON
- h) It shall not be possible to close the breaker under following conditions:
- Stored energy mechanism is not charged
 - Shunt release is energized
 - OFF push button is locked in OFF position
- i) The circuit breaker shall be provided with mechanical ON/OFF, 'TRIP', 'SPRING CHARGED/DISCHARGED', 'READY TO CLOSE' and breaker position indications and mechanical trip push button. An operating handle shall also be provided to charge the spring manually.
- j) The circuit breaker shall be provided with automatic safety shutters, so that before the breaker reaches 'ISOLATED' position the main isolating contacts are completely shrouded. In the drawout condition, it shall be possible to inspect the breaker fixed contacts condition by lifting the shutters.
- k) When the circuit breaker compartment door is open, it shall not be possible to touch the live parts. All removable covers protecting live parts shall be clearly labeled with warning notices reading 'LIVE PARTS. ISOLATE ELSEWHERE BEFORE REMOVING COVER'. It shall be possible to readily remove the arc chutes for routine inspection of the contacts with the circuit breaker in the withdrawn position. Contact erosion indicator shall be provided on moving contacts of breaker for visual indication of contact life.
- l) The circuit breaker compartment shall be so designed that hot gases produced during fault shall be let away from the operator.
- m) The protective relays and instruments shall be mounted in separate compartment. This compartment shall be along side and immediately next or above to the breaker.
- n) All the non-conducting metal parts of the circuit breaker trolley shall be bonded together and shall make perfect electrical connection to earth through substantial sliding contacts at 'SERVICE' and 'TEST' positions. Such sliding contacts shall be arranged to make before power plug-in and interrupt after power draw-out.
- o) Circuit breaker operation shall be independent of the motor, which shall be used only for charging of the spring. The mechanism shall preferably be such that:
- With the circuit breaker open and the springs charged, the circuit breaker can be closed and tripped
 - With the circuit breaker closed and the springs charged, there shall be sufficient energy to trip, close and then trip

Tripping and closing circuits shall be provided with miniature circuit breaker in each pole on each unit and shall be independent of each other and all other circuits.

- p) Each breaker feeder shall be provided with the following as a minimum:

- An anti-pumping relay
 - Motor charged spring operating mechanism
 - Manual spring charging and manual closing mechanism
 - Mechanical indication of spring charge
 - Mechanical position indicator (to show whether the breaker is in the 'ON' or 'OFF' positions and in the 'SERVICE', 'TEST' or 'ISOLATED' positions)
 - Closing coil
 - Shunt trip coil
 - CTs for protection
 - Manual close and trip push button
 - Phase barriers
 - Shutter assembly
 - Door interlock kit
 - Operation counter
- q) The closing coil and spring charging motor shall operate satisfactorily for all values of control supply voltage between 80% and 110% of the rated voltage. But, the shunt trip coil shall be designed such that it will operate satisfactorily for all values of control supply voltage between 70% and 110% of the rated voltage. One open-close-open operation of the circuit breaker shall be possible after failure of power supply to motor.
- r) Spring charging motor shall be provided with overcurrent protection. Motor windings shall be provided with class B insulation or better.
- s) Following is a list of minimum circuit breaker malfunctions that shall be prevented by the use of mechanical interlocking devices:
- The circuit breaker being withdrawn from or inserted into the primary isolating contacts when the circuit breaker is closed or the plug for secondary disconnection, where provided, is not fully engaged
 - The closing of the circuit breaker unless correctly located in the 'SERVICE', 'ISOLATED' or 'TEST' position
 - The circuit-breaker being closed in the 'SERVICE' position without completing the auxiliary circuits via the secondary isolating devices, where self-aligning secondary isolating contacts have not been provided
 - The breaker cubicle door being opened unless the breaker is in the 'OFF' or 'ISOLATED' condition
 - Closing of circuit breaker in 'SERVICE' position with the door open
 - Removal of the plug for secondary disconnection when the breaker is in 'SERVICE' position and is ON

Locking facilities shall include the following:

- Circuit breaker control switch in the neutral position
- Mechanical trip on circuit breakers so as to prevent manual trip
- Access flaps for operating or withdrawing of circuit breakers

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Breakers controlling motors shall operate satisfactorily under the following conditions:

- Direct-on-line starting of induction motors rated above 75kW with a locked rotor current of 5.6 times the rated current with 20% tolerance and starting time of upto 30 seconds
 - Breaking no load, full load and locked rotor current of induction motors as stated above
 - No harmful overvoltage shall be produced during switching off of induction motors. If required, surge protective device shall be provided to limit over voltages
- t) Telescopic trolley or suitable arrangement shall be provided for maintenance of circuit-breaker module in a cubicle. The trolley shall be such that the topmost breaker module can be withdrawn on the trolley and can be lowered for maintenance purpose. Also, it shall be such that all types, sizes and ratings of breakers can be withdrawn/inserted in the switchboard

203.6 **Contactor Starter Unit**

- a) All motors rated up to 75kW shall be controlled by direct-on-line contactor starters. MPCB shall be used for motor feeders rated 10kW and below. For feeders rated above 10 kW and upto 75kW, MCCB shall be used. The starting circuit shall consist of the following:



- Isolating device (MCCB/MPCB)
 - Contactor as main means of starting and stopping of motor
 - A short circuit protective device (Inbuilt feature of MPCB/MCCB)
 - Electronic overload protection relay with current display unit & built-in single phasing prevention feature
 - CT operated numerical relay for motor feeders rated 75kW and above
- b) Withdrawable contactor starter units shall be provided with means for mechanically indicating the ‘SERVICE’ and other positions.
- c) The starter units shall carry designation labels in terms of module type codes, signifying the function of the unit. Labels indicating these shall be affixed on fixed and withdrawable portion of the unit.
- d) The units shall be designed to ensure safety of operating personnel. Interlocks shall be provided to ensure that the unit access door can only be opened when the associated isolating device is open. Defeat interlock facility shall be provided.
- e) The starters shall be readily interchangeable with each other. Combination starters shall be drawout.
- f) Electronic overload relays shall have reset push button.

203.7 Moulded Case Circuit Breaker (MCCB) / Motor Protection Circuit Breaker (MPCB)

MCCBs shall be of three / four pole construction for panel mounting. The MCCBs shall be provided with front operating handles and mechanical ON / OFF indicators.

The Moulded Case Circuit breaker shall be provided with spring assisted quick-make, quick-break, current limiting type manually operated trip free mechanism, mechanical ON/OFF/TRIP position indicators, thermal tripping devices of inverse characteristics, instantaneous short circuit tripping devices and necessary auxiliary and alarm contacts. The MCCB module shall be provided with service, test and isolated position. The thermal and short circuit tripping device shall be adjustable type.

The MPCB's shall be in generally similar to that of MCCB's in all the features mentioned above.

MCCBs/MPCBs shall be provided with overload thermal release setting range of 50% to 100% of rated current and adjustable short circuit magnetic release of 5 to 10 times rated current.

MCCBs/MPCBs shall be capable of withstanding the thermal stresses caused by overloads and locked rotor currents of values associated with protective relay settings of the motor starting equipment and the mechanical stress caused by the peak short circuit current of value associated with the switchgear rating. The maximum tripping time under short circuit shall not exceed 20 ms. When used for motor circuit, shunt trip devices shall be provided and the let through power of controlling MCCB/MPCB shall be lower than the respective contactor, Contactor and overload relays shall be selected so as to withstand the let-through energy of the connected MCCB/MPCB in the feeder and consequent thermal and dynamic effects. All power feeder module MCCBs shall be equipped with earth fault release.

The DC circuits shall be provided with DC MCCB / MCBs.
MCCBs/MPCBs shall have following accessories and features:

- a) Shunt trip release
- b) Auxiliary contact set of 1 NO + 1 NC
- c) Fault signaling contact set of 1 NO + 1 NC
- d) Insulation shields to isolate the connection between each pole
- e) Finger protection plate to prevent accidental contact

The compartment door shall be interlocked with handle of MCCB/MPCB

203.8 Control and Selector Switches

- a) All circuit breaker operating switches shall be of the pistol grip type, spring return to neutral and lockable in that position. They shall be arranged to close the breaker by being turned clockwise and to trip it by being turned anti-clockwise. The trip, neutral and close positions shall be clearly indicated. The movement shall be such that the switch cannot be operated inadvertently and that it is mechanically

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interlocked to trip before close. The control switch shall be located at a suitable height for ease of operation.

- b) Ammeter and voltmeter selector switches shall have four stay put positions with adequate number of contacts for 3 phase 4 wire system. These shall have oval handles. Ammeter selector switches shall have make before break type contacts to prevent open circuiting of CT secondary. The arrangement for front mounting of these devices shall be such as to make them reasonably dust free so as not to interfere with normal operation.
- c) Local/Remote selector switch shall be provided in switchboard

203.9 Contactors

- a) Motor starter contactors shall be of air break, electromagnetic type rated for uninterrupted duty. Contactors shall be double-break, non-gravity type and their main contacts shall be silver faced copper.
- b) Direct-on-line contactors shall be of utilization category AC3. Full voltage reversing starters shall comprise of forward and reverse contactors mechanically and electrically interlocked with each other. These contactors shall be of utilization category AC4. DC contactor shall be of utilization category DC3.
- c) All springs shall be made out of corrosion-proof material.
- d) The number of NO and NC auxiliary contacts of a contactor shall be as per requirement shown in the respective module drawings. It shall, however, be not less than 2 NO+2 NC.
- e) The contactor shall operate satisfactorily between 80% and 110% of the rated voltage. The contactor shall not drop out at 70% of the rated voltage but shall definitely drop out at 20% of the rated voltage.
- f) The contactor rating shall be chosen to provide type-2 discrimination. However, the minimum rating shall be 16 A.

203.10 Miniature Circuit Breaker (MCB)

- a) MCB shall be rated for 10kA short circuit rating. It shall be quick make, quick break, independent manual type with trip free feature. MCB shall have the following :
 - Overcurrent protection
 - ON/OFF position indicators
 - Auxiliary contact block for ON/OFF/Trip positions

203.11 Instrument Transformers

All current and voltage transformers shall be completely encapsulated, cast resin insulated type suitable for continuous operation at the temperature prevailing

inside the switchgear enclosure, when the switchboard is operating at its rated condition and the specified ambient temperature. The class of insulation shall be 'B' or better.

All instrument transformers shall be able to withstand the thermal and mechanical stresses resulting from the maximum rms short circuit breaking and peak making current ratings of the associated switchgear.

Current transformers may be multi or single-core type. All voltage transformers shall be single phase type. Accuracy class & VA burden shall be as specified in single line diagram.

The bus VTs shall be housed in a separate compartment. All VTs shall have readily accessible MCCB and MCBs on primary and secondary sides respectively.

203.11.1 Current Transformer

- a) Current Transformers shall be cast-resin type with 1A secondary. Separate cores shall be used for metering and protection. All secondary connections shall be brought out to terminal blocks where wye or delta connection will be made. Shorting links shall be provided for the secondary of the CTs when wired to terminals. CT secondary terminals shall be easily accessible and shall be provided on front LT compartment. One side of CT secondary shall be earthed in the compartment with the meters or relays which they serve and each CT group shall be earthed with a separate identified lead which may be disconnected for testing.
- b) Accuracy class of the current transformers shall be:
 - i) Class 5P20 & PX for protection
 - ii) Class 0.5 and ISF<5 for metering

203.11.2 Voltage Transformer

- a) Voltage transformers shall be cast-resin drawout type and shall have an accuracy class of 0.5/3P for metering & protection. Voltage transformer mounted on breaker carriage is not acceptable.
- b) The burden of VT shall be 50VA for protection core and 100VA for metering core. The primary of the VT shall be rated for 400 volts and the secondary for 100 volts. It shall have a continuous voltage factor of 1.2 with short time over voltage factor of 1.5 for 30 seconds. The VT shall be provided with MCCB on the primary side and MCBs on secondary side.
- c) The voltage transformer shall be completely disconnected and visibly grounded in fully drawout position.
- d) For proper relaying, one side of VT secondary shall be earthed at the transformer and the earth connection shall be identified and removable for testing. Test terminals shall be provided for VT circuits.

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203.12 Relays and Timers

Relays and timers in protective circuits shall be flush mounted on panel front with connections from the inside. They shall have transparent and dust tight covers removable from the front. They shall either have built-in test facilities, or shall be provided with necessary test blocks and test switches located immediately below each relay.

Relays shall have lock-out facility with manual reset. Each circuit breaker feeder shall be provided with separate latched lockout relay (86), trip circuit supervision relay, necessary auxiliary relays, timers, etc. to meet circuit requirements. In addition self-reset type lockout relay to be provided on Bus PT Panel for Under voltage tripping the motor feeders.

Lock out relay contacts shall only be provided for protection trip & reset status for use in close and trip interlocks of the circuit breaker.

Auxiliary supply voltage for relays and transducers shall be 125V DC. Relays shall be capable of satisfactory continuous operation between 80-120% of the rated voltage.

Failure of a control or auxiliary supply and de-energisation of a relay shall not initiate any circuit breaker operation. All relays shall withstand a minimum test voltage of 2.5 kV AC RMS for one minute.

The relays shall have provision of both current and voltage inputs. The current operated relay shall have provision for 5 sets of CT inputs, 3 nos. for phase fault and 1 CT input for earth fault & 1 CT input for CBCT. Relay shall be suitable for both residually connected CT input as well as CBCT input. Relays shall be suitable for CT secondary current of 1A/5A selectable at site. The voltage-operated relay shall have provision for 3 PT inputs. Relays shall be rated for operation on 100V VT secondary voltage. Relays used in Incomers and bus section breaker shall have provision of two sets of voltage signal inputs for the purpose of synchronization.

All CT and PT terminals shall be provided as fixed type terminals on the relay to avoid any hazard due to loose connection leading to CT opening or any other loose connection. In no circumstances Plug In type connectors shall be used for CT/ PT connections.

All numerical relays shall have key pad / keys to allow relay settings from relay front. All hand reset relays shall have reset button on the relay front. Relay to be self or hand reset shall be software selectable. Manual resetting shall be possible from remote. Relays shall have suitable output contact for breaker failure protection.

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Relays shall have self-diagnostic feature with self-check for power failure, programmable routines, memory and main CPU failures.

The numerical relay shall be able to provide supervisory functions such as, circuit breaker state monitoring, PT and CT supervisions and recording facilities with Post fault analysis

The numerical processor shall be capable of measuring and storing values of a wide range of quantities, all events, faults and disturbance recordings with a time stamping using the internal real time clock. Battery backup for real time clock in the event of power supply failure shall be provided.

Relays shall have event recording feature, recording of abnormalities and operating parameters with time stamping.

Numerical relays shall be capable of storing Minimum of 150 time tagged events/ records with time stamping. The numerical relays shall be able to store last 5 faults including the indication, waveforms, protection operated, fault location relay and operating time, currents, voltage and time. All Setting parameters, Fault data, waveforms & event logs shall be stored in Non-volatile memory only.

Sequence of events shall have 1ms resolution at device level.

Measurement accuracy shall be 1 % for RMS Current and voltage.

Relay shall be immune to capacitance effect due to long length of connected control cables. Any external hardware, if required for avoiding mal operation of the relay due to cable capacitance shall be included as a standard feature. All IOs shall have optical isolation. Analog inputs shall be protected against switching surges, harmonics etc.

The alarm/status of each individual protection function and trip operation shall be communicated to PCS/PLC/PC. The numerical relay system shall have built-in features/hardware interface to provide such inputs to PLC for analog/digital values.

Relays shall have at least two sets or groups of two different sets of adaptable settings. Relays shall have multiple IEC/ ANSI programmable characteristics.

Numerical relays shall have two level pass word protections, one for read only and other for authorization for modifying the setting etc.

Timer functions shall be programmable for on/off delays.

The protective relays shall have at least 10 nos. potential free contacts (Programmable) Auxiliary relays shall have contacts as required. Relay output contacts shall be suitable for directly wiring in the breaker closing and trip circuit operating from 125V DC control voltage.

Making, carrying and breaking current ratings of their contacts shall be adequate for the circuits in which they are used. Interrogation voltage for the binary inputs shall be suitably selected to ensure avoidance of mal operation due to stray voltages.

No separate earth bus shall be required for the relays. It shall be possible to connect the relay earth to the common earth bus in the switchgear panel which shall be connected to the plant earth mat

All protective relays shall be latest numerical type, having following features/functions:

- Protection functions as required
- IEC IDMT characteristics
- Measurements
- Event/fault/disturbance recording
- Binary input/output relays as required
- LEDs to indicate status of the relay
- LCD display for measurements, settings, faults etc
- Key pad on the front of the relay
- Communication ports for transmitting settings, measurements, alarms, faults, events, disturbance records to PCS/PLC/SAS (Substation Automation System)
- Communication port for software setting / access measurements / maintenance / fault analysis
- Self-monitoring

All protective relays shall be latest numerical type with one USB port at the front for connecting to Laptop and one port (RJ45/FO port) at rear suitable for communication on MODBUS TCP/IP.

The Numerical relays (MV/LV System) shall be networked through Ethernet switch, further integrated with Plant Control System (PCS) through MODBUS TCP/IP communication system (for monitoring, measuring, fault data analysis & relay parameterization).



All necessary hardware including Managed Ethernet switches, accessories and licensed software shall be supplied by the vendor.

Ethernet switches shall be 'substation hardened' and shall comply with IEC61850-3 for communications and environment requirements. The Ethernet switches shall be of managed type with two (2) no. of fibre optic cable ports for ring network and Fourteen/Sixteen of Copper ports to achieve the LAN configuration. These switches shall be mounted inside the switchgear Panel.

The switchboard shall be provided with DC fail relay and DC fail indication lamp.

Breaker auxiliary contacts used for interlocking purposes shall be multiplied using electrically latched relay.

Protection relays shall be provided as detailed in the Protection single line diagrams. The protection shall include, protection functions but not limited to the following:

- a) Incomer feeder from transformers:
- Instantaneous over current protection (50)
 - IDMT over current protection (51)
 - Instantaneous Earth fault protection (50N)
 - IDMT Earth fault protections (51N)
 - Under voltage relay (27)
 - Check synchronizing relay (25C)
 - Standby Earth fault protections through Transformer Neutral CT (51N1)
 - Lockout Relay / Master trip Relay (86)
 - Anti-Pump Relay (94)
 - C.B Trip coil supervision (95)
 - Fuse Failure Relay (98)
- b) Incomer feeder from BSDG:
- Current operated differential relay (87)
 - O/C relays (50/51) for phase fault with timer (2)
 - O/C relay (50N/51N) for Earth fault with timer (2).
 - Under voltage relay (27)
 - Standby earth fault relay (51N1)
 - Breaker fail protection relay (50LBB)
 - Lockout Relay / Master trip Relay (86)
 - Anti Pump Relay (94)
 - C.B Trip coil supervision (95)
 - Fuse Failure Relay (98)
 - Check Synchronizing relay (25C)
- c) Normal Incomer feeder (Direct from Upstream switchboard):
- Instantaneous over current protection (50)
 - IDMT over current protection (51)
 - Instantaneous Earth fault protections (50N)
 - IDMT Earth fault protections (51N)
 - Check Synchronizing Relay (25C)
 - Lockout Relay/Master Trip Relay (86)
 - Anti Pump Relay (94)
 - C.B Trip coil supervision (95)
 - Fuse Failure Relay (98)
- d) Bus Section Breaker feeder (ACB):
- Instantaneous Overcurrent Protection (50)
 - IDMT Overcurrent Protection (51)
 - Instantaneous Earth fault protections (50N)
 - IDMT Earth Fault Protection (51N)
 - Local Breaker Backup Protection (50LBB)



- Check Synchronizing Relay (25C) for PCC/MCC only.
 - Lockout Relay/Master Trip Relay (86)
 - Anti-Pump Relay (94)
 - C.B Trip coil supervision (95)
- e) Outgoing ACB operated Motor feeder:
- O/C relays (50/51) for phase fault with timer (2)
 - O/C relay (50N/51N) for Earth fault with timer (2)
 - Comprehensive Motor protection relay with feature not limited that mentioned below:
 1. Thermal Over load (49)
 2. Negative sequence (46)
 3. Phase Reversal (46 R)
 4. Restart inhibit with startup counter (37)
 5. Locked rotor (48-LR)
 6. Under current (66)
 - Lockout Relay (86)
 - Anti Pump Relay (94)
- f) Other Outgoing feeder:
- O/C relays (50/51) for phase fault with timer (2)
 - O/C relay (50N/51N) for Earth fault with timer (2)
 - Lockout Relay (86)
- g) Bus PT:
- 3 phase Under voltage relay
 - Fuse Failure Relay (98)
 - Lockout Relay (86)

203.13 Indicating Instruments

All indicating instruments and meters shall be capable of carrying continuously their full load currents and full voltage across their coils. They shall not be damaged by the passage of fault currents or the existence of over voltage on the primary side of their instrument transformers for the maximum permitted duration of fault conditions which may occur during normal operation. All instruments and meters shall be back connected.

- a) Meters and transducers for 400V PCCs
- For incomers feeders following Meters and transducers shall be provided:
- Current transducer on 3 phases
 - Voltage transducer on 3 phases
 - 0.5 class multifunction meter with three phase current, three phase voltage, Frequency, kVA, kW, kVAr, kWh and power factor
- For Bus Section Breaker following Meters and transducers shall be provided:
- Current transducer on 3 phases
 - Voltage transducer on 3 phases
 - 0.5 class Ammeter

For ACB outgoing feeders following Meters and transducers shall be provided:

- Current transducer on 'Y' phase
- 0.5 class multifunction meter with three phase current, three phase voltage, Frequency, kVA, kW, kVAr, kWh and power factor.

For ACB motor feeders following Meters and transducers shall be provided:

- Current transducer on 'Y' phase
- 0.5 class Ammeter

For contactor motor feeders 10kW & above following Meters and transducers shall be provided:

- Current transducer on 'Y' phase
- 0.5 class Ammeter
- Electronic overload relay with current display.

For contactor motor feeders less than 10kW following Meters and transducers shall be provided:

- Electronic overload relay with current display

For Bus PT following Meters and transducers shall be provided:

- 0.5 class Digital Voltmeter measuring all three phases voltage & frequency
- Voltage transducer on three phases

b) Meters and Transducers for 400V MCCs

For incomers feeders following Meters and transducers shall be provided:

- Current transducer on 3 phases
- Voltage transducer on 3 phases
- 0.5 class digital multi-function Meter with three phase current, three phase voltage, Frequency, kVA, kW, kVAr, kWh and power factor

For Bus Section Breaker following Meters and transducers shall be provided:

- Current transducer on 3 phases
- Voltage transducer on 3 phases
- 0.5 class Ammeter

For contactor motor feeders 10kW & above following Meters and transducers shall be provided:

- Current transducer on 'Y' phase
- 0.5 class Ammeter
- Electronic overload relay with current display.

For contactor motor feeders less than 10kW following Meters and transducers shall be provided:

- Electronic overload relay with current display

For Bus PT following Meters and transducers shall be provided:

- 0.5 class Digital Voltmeter measuring all three phases

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- Voltage transducer on three phases
- c) Meters and Transducers for 400V DBs

For incomers feeders following Meters shall be provided:

- 0.5 class Digital voltmeter measuring all three phases
- 0.5 class Digital ammeter measuring all three phases

For Bus PT following Meters shall be provided:

- 0.5 class Digital Voltmeter measuring all three phases

- d) Multifunction Meter shall be provided with LCD display for local indication and shall be provided with communication port suitable for communicating on MODBUS protocol.
- e) Multifunction Meters shall display various parameters like 3 phase currents, voltages, PF, MD, KW, KVA, Hz, harmonics, etc. for local indication and shall be provided with minimum 8 nos. of user programmable registers for communicating any of the displayed parameters to the remote. Multifunction Meters shall be looped in Daisy Chain fashion with maximum of 10 meters connected in a loop. Each loop shall be connected to Modbus over TCP/IP protocol converter further integrated with PCS through Ethernet Switch / Data Concentrator.
- f) Analog indicating meter shall have Accuracy class 0.5 and shall be 96x96mm square with 240° scale. Ammeters of motor feeders shall have cramped scale above 2 times to 6 times the rated current for indicating the starting current.
- g) Transducers shall be of dual output type, class 0.5, 4-20mA DC linear output & 750 ohm load.

203.14 **Push Buttons**

- a) Push buttons shall be of heavy duty, spring return and push-to-actuate type. Their contacts shall be rated to make, continuously carry and break 10A at 230V AC and 1A (inductive) at 125V DC
- b) All push buttons shall have two nos. NO and two nos. NC contacts, unless specified otherwise. The contact faces shall be of silver alloy. All push buttons shall be provided with integral escutcheon plates marked with their functions.
- c) The colour of the button shall be as follows:
- Green for motor START and breaker CLOSE
 - Red for motor STOP and breaker OPEN
 - Black for valve/damper OPEN/CLOSE commands
 - Black for all annunciation functions, overloads, reset and miscellaneous commands

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- d) All push buttons on panels shall be located in such a way that red push buttons shall always be to the left of green push buttons. All emergency push buttons shall have mushroom knobs with protective glass.

203.15 **Indicating Lamps**

- a) Indicating lamps shall be of the panel mounting clustered LED type. The lamps shall have escutcheon plates marked with their functions, wherever necessary. Bulbs and lamp covers shall be easily replaceable from the front of the cubicles. Low Voltage Glow Prevention (LVGP) feature shall be provided for indicating lamps. Lamps shall have translucent lamp covers of the following colours, as warranted by the application :
- Red for motor ON, valve/damper OPEN and breaker CLOSE
 - Green for motor OFF, valve/damper CLOSE and breaker OPEN
 - Amber for TRIP
 - Blue for 'SERVICE' position indications
 - White for all healthy conditions (e.g. control supply, lockout relay healthy, spring charged condition etc.)
 - White for 'TEST' position indications
 - Red, yellow, blue for R, Y and B phases respectively
- b) All indicating lamps shall be suitable for continuous operation at 85% to 110% of their rated voltage.
- c) For incomer and bus, indicating lamps for R, Y and B phases shall be provided

203.16 **Secondary MCBs**

- a) Secondary circuit shall be protected by a set of MCBs as per requirements of schemes.
- b) MCBs shall be provided with thermal overload and magnetic short circuit protection. MCBs shall be provided with auxiliary contacts for ON/OFF/Trip conditions. MCBs shall be selected for correct discrimination and adequate ratings. The protection tripping of these devices shall be adequately guaranteed when used in low fault level circuits viz. on the secondary of VTs.

203.17 **Wiring**

- a) Wiring shall be by 650 volt grade multi-stranded PVC insulated copper wire having a cross-sectional area of not less than 1.5sq.mm. All connections from CT leads upto instruments, relays and terminal board shall be made by copper wires of minimum 2.5sq.mm size. The cables shall be tested for flammability test as per applicable standards and shall also withstand service temperature without deterioration.

Internal panel copper wiring requirement will be as follows:

- a) Cables for CT/PT secondary wiring : 2.5sq.mm, brown, black, grey and blue for L1,L2,L3 and Neutral respectively.
- b) Cables for auxiliary AC circuit : 2.5sq.mm, brown for Phase wire and blue for Neutral wire.
- c) Cables for DC Control circuit : 1.5sq.mm, brown for positive wire and blue for negative wire.
- d) Potential free circuit : 1.5sq.mm, grey.
- e) Shield cable : 1sq.mm, 2 core shielded.
- f) Cables for earthing : 2.5Sq.mm, yellow with green strip line.
- g) Cables for annunciator : 1.5Sq.mm, white
- b) Wiring shall be run mostly clear of all metal parts in insulated cleats, properly routed and neatly bunched. However, PVC wire holders and channels shall be preferred for running of wiring. Where wiring passes from one compartment to another, the aperture shall be 'bushed' to prevent damage of wires against sheet metal edges. Bushes may comprise of good quality rubber grommets.
- c) Where single phase conductors are associated with the 3 phase system from which they are derived, the phase conductor shall use the same colour as the phase from which it is derived.
- d) Each Shipping Section of Switchgear shall be provided with Seperate Marchalling Compartment for DCS/PCS Interface wiring. Internal Wiring shall be done through Troughs inside Switchgear from feeders of the shipping Section to Marshalling Compartment.
- e) Inter-module bus wires shall be kept separate from all other wiring. AC or DC terminations shall be grouped function-wise as far as possible and labels of the function shall be affixed.
- f) Terminal boards shall have separate terminals for incoming and outgoing wires with not more than two wires connected to any one terminal. Terminal boards shall be mounted vertically at the side of the cubicle or in horizontal rows and properly spaced to have clean wiring arrangement, adequate access for putting ferrules, making terminations etc. It shall be possible to read the ferrule numbers when the wiring is complete. Where terminals may be live when the equipment is isolated from the main supply, these shall be clearly marked on the panel.
- g) Extra flexible wires shall be used for wiring to devices mounted on moving parts such as hinged doors. The wire bunches from the panel inside to the doors shall be properly sleeved or taped. All internal wiring terminations shall be made with solderless crimping type heavy duty tinned copper lugs. Insulation sleeves shall be provided over the exposed parts of lugs.
- h) Engraved core identification plastic ferrules marked to correspond with panel wiring diagrams shall be fitted at both ends of each wire.



- i) Control terminal blocks shall be of 1100V grade, rated for 10A and in one piece molding. It shall be complete with insulating barriers, clip-on type terminals and identification strips. Marking on terminal strip shall correspond to the terminal numbering on wiring diagrams. Terminal blocks for CT and VT secondary leads shall be provided with test links and isolating facilities. CT secondary leads shall be provided with short circuiting and earthing facilities.
- j) In all the panels, at least 20% spare terminals for external connections shall be provided and these spare terminals shall be uniformly distributed on all terminal blocks.
- k) All interlocks shall be hardwired in the switchboard itself.
- l) The wiring shall be complete in all aspects so as to ensure proper functioning of control, protection and interlocking scheme. All the wiring shall be complete up to terminal blocks on the side of each unit/module.
- m) Adjacent rows of terminal blocks shall be spaced not less than 100mm apart.
- n) The terminal blocks shall be mounted inside the module (and not in cable way) in a manner so as to provide easy access to terminals and to enable ferrule numbers to be read without difficulty.
- o) All inter-panel wiring shall be carried out within the switchboard before dispatching the panel.
- p) A typed circuit directory, card holder and card with clear plastic covering shall be provided on the inside of each cabinet door. The directory card shall provide space to identify each circuit in the panel board. An electronic copy of the circuit directory shall be provided to the Purchaser.

203.18 **Cable / Bus duct Termination**

- a) PMCC/MCC/DB shall be designed for cable entry from the bottom. Sufficient space shall be provided for ease of termination and connection. Add-on panels shall be considered for cables, if any.
- b) Cable termination compartment and arrangement for power cables shall be suitable for heavy duty, 1.1 kV grade, stranded copper conductor, XLPE insulated, un-armoured and FRLS PVC sheathed cables.
- c) Sufficient space and support arrangement shall be provided in the cubicles to accommodate cables.
- d) Control Cables shall be PVC insulated, un-armoured, extruded FRLS PVC overall sheathed.
- e) All provisions and accessories shall be furnished for termination and connection of cables, including removable gland plates, cable supports, crimp type tinned copper lugs, brass double compression glands with tapped washer and terminals blocks.
- f) The gland plates and supporting arrangement for single core power cable shall be such as to minimize flow of eddy current.

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203.19 **Name plates and Labels**

- a) Nameplates shall be of 3 ply laminate with black lettering on white background. The following shall be provided with nameplates/warning labels as described below:

- Main nameplate for PCC/MCC/DB on front and back sides
- All incomers and outgoing feeders indicating description, rating, equipment no., feeder no., etc
- All door mounted components
- Panel numbers on front and rear
- Warning labels for interlocks
- Incoming supply and outgoing supply tagging including KKS

All nameplates shall be fastened by means of screws to the panel.

Danger labels shall be provided for the following:

- MCC as per statutory regulations
 - Bus bar chamber
 - Cable alley housing live terminals
- b) Nameplate or polyester adhesive stickers shall be provided inside the panel for all equipment (lamps, push buttons, switches, relays, auxiliary contactors etc.) mounted on the switchboard.
- c) Special warning plates, one each on each front of a shipping section, shall be provided on removable covers of doors giving access to cable terminals and bus bars. Special warning labels shall be provided inside the switchboard also, wherever considered necessary. Identification tags shall be provided inside the panels matching with those shown on the circuit diagram.
- d) All major components such as breakers, switches, contactors, CTs etc. shall be provided with labels indicating their ratings.
- e) For single front switchboards, similar panel and board identification labels shall be provided at the rear also.

203.20 **Earthing**

- a) Two earth terminals shall be provided on each switch cubicle, at the rear side. An earth bar of at least 50x6 mm copper flat shall be fixed to these terminals. The earth bar shall be electrically continuous and shall run the full extent of each board. This earth bar shall be on the same side as the cable entry and shall be extended on both sides of the panel.
- b) Each unit shall be constructed to ensure satisfactory electrical continuity between all metal parts not intended to be alive and earth terminals of the unit. Suitable holes with bolts and nuts shall be provided at each end of earth bar of switchboard for connection to a main earthing grid.

- c) The earth bar shall be accessible in each cable entering compartment either directly or through a branch extension to earth the cable armour and shields. 10mm diameter holes shall be drilled and hardware for connection provided through the earth bus.
- d) The cores of transformer (bus voltage transformer, control transformer, heating transformer etc.) shall also be earthed through copper wire. Doors shall have a copper wire for earth connection to fixed unit. VT and CT secondary neutral point earthing shall be at one place only on the terminal block.
- e) All non-current carrying metal work of the switchboard/panel shall be effectively bonded to the earth bus. All metallic cases of relays, instruments and other panel mounted equipment shall be connected to earth bus by independent stranded copper wires of size not less than 2.5sq.mm. Insulation colour code of earthing wires shall be green. Earthing wires shall be connected to terminals with suitable clamp connectors and soldering shall not be acceptable. All hinged doors shall be earthed through flexible earthing braid.

203.21 **Local Motor Starters**

- a) Local motor starters shall be metal enclosed, suitable for outdoor mounting on wall or steel structures with suitable hood on top. The enclosure shall be cold rolled sheet steel of at least 1.6mm thickness. The local motor starters shall be dust and vermin-proof and shall have a degree of protection of IP-55W.
- b) Local motor starters shall comprise of the following:
 - A 3-pole contactor, mechanically latched type
 - Start push button, coloured green
 - Stop push button, coloured red
 - 'ON' indicating lamp
 - Ambient temperature compensated thermal overload relay with single phasing protection

203.22 **Local Push Button Stations (LPBS) (Optional)**

- a) The local push buttons stations shall be FRP/flame-proof enclosure with hood on top suitable for outdoor mounting on wall or steel structures. The enclosure shall be provided with a protective guard, to avoid inadvertent operation of push buttons. The local push button stations shall be dust and vermin-proof and shall have a degree of protection of IP-55W for outdoor and IP 52 for indoor as per IEC 60529
- b) LPB shall be of following types:
 - Type-A: With Stop & Mushroom Head type Emergency Stop Push Button in FRP enclosure
 - Type-B: With Start, Stop & Mushroom Head type Emergency Stop Push Buttons in FRP enclosure
 - Type-C: With Forward, Reverse, Stop & Mushroom Head type Emergency Stop Push Buttons in FRP enclosure
 - Type-D: With Stop & Mushroom Head type Emergency Stop Push Button in

- flame-proof enclosure
 - Type-E: With Start, Stop & Mushroom Head type Emergency Stop Push Buttons in flame-proof enclosure
 - Type-F: With Forward, Reverse, Stop & Mushroom Head type Emergency Stop Push Buttons in flame-proof enclosure
- c) Push buttons shall be with two NO and two NC contacts. Emergency stop push buttons shall be mushroom, latched type. Start push button shall be self reset type.

203.23 **Painting**

The final coat shall be of a glossy oil and weather resisting non-fading paint of shade number RAL 7032. Please note that site is near the sea. Painting of the equipment should be selected accordingly. Painting selected shall give a minimum life of 20 years to first maintenance. Surfaces which are inaccessible after erection shall be protected for a 25 years life. The procedure of Corrosion Protection by Protective Paint Systems will be in accordance to Corrosive Environment C4 H for indoor and Durability of the coating system as High Durability >15 years design life (first major maintenance) in compliance with ISO 12944. Primer paint shall be ready made zinc chrome. The final thickness of paint film on steel shall not be less than 80 microns. Sufficient quantity of touch-up paint shall be furnished for application at site.



Any damage to paint work incurred during transport and erection shall be made good by the Vendor by thoroughly cleaning the damage portion and applying the full number of coats of paint that had been applied before the damage was caused. One coat of additional paint shall be given at site over all external surfaces, including hardware, after erection. Vendor shall supply the requisite quantity of paint.

203.24 **Tests**

- a) All tests shall be conducted as per relevant IEC standards and shall be performed in the presence of Purchaser's representative, if so desired by the Purchaser. The Vendor shall give at least 15 days advance notice of the date when the tests are to be carried out.
- b) All equipment/components/materials shall be of type tested and proven type. Relevant type test reports shall be submitted to the Purchaser for approval. Type tests should have been carried out within the last five years as on the date of opening of the bid on equipment of similar rating and on identical components/materials.
- c) For the various bought out items, test certificates from the manufacturers shall be furnished.
- d) Copies of certified reports of all tests carried out at the works shall be submitted. The equipment shall be dispatched from works only after receipt of Purchaser's written approval of the test reports

203.25 **Type Tests**

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- a) Type test reports for the following shall be submitted for switchgear panel of each current rating :
- Power frequency withstand test on power circuit and auxiliary and control circuits
 - Temperature rise test
 - Measurement of resistance of the main circuit
 - Short time withstand current and peak withstand current test
 - Degree of protection test
- b) Type test reports shall be submitted for circuit breaker as per applicable standard by the Bidder for Purchaser's review. The list of type tests shall include the following:
- Dielectric tests
 - Measurement of the resistance of the main circuit
 - Temperature rise tests
 - Short time withstand current and peak withstand current tests
 - Electromagnetic compatibility tests
 - Mechanical and environmental tests including mechanical operation test
 - Short circuit current making and breaking tests
 - Mechanical endurance test
 - Electrical endurance test
- c) The contactors, relays, switches, breakers, local push buttons, local motor starters, instruments, VTs, CTs and control transformers shall be subjected to type tests in accordance with the relevant IEC standards

203.26 **Routine Tests**

- a) The switchgear panel shall be subjected to routine tests in accordance with the relevant IEC standards. The tests shall include the following:
- Power frequency voltage dry test on the main circuit
 - Voltage test on control and auxiliary circuits
 - Measurement of resistance of the main circuit
 - Mechanical operating tests
- b) The circuit breakers, contactors, switches, local push buttons, local motor starters, VTs, CTs and control transformers shall be subjected to routine tests in accordance with the relevant IS/IEC standards. Routine test reports shall be furnished by the Vendor for Purchaser's review.
- c) For circuit breaker, the list of routine tests shall include the following:
- Dielectric tests on the main circuit
 - Dielectric tests on the control circuit & auxiliary circuit
 - Measurement of the resistance of the main circuit
 - Mechanical operating tests
 - Design and visual checks

203.27 DI/DO/AI Signal interface with PCS



- a) For each incomer breaker, Tie-breaker and outgoing feeders, the following signal interface shall be considered.

S. No.	Description Feedback	Type of I/O	No of I/O	Remarks
	General		For PCS	
1	Incomer		For PCS	
a	Breaker CLOSED Feedback	DI	1	Hardwired To PCS
c	Breaker test position	DI	1	
d	Breaker available / Trip	DI	1	
e	Breaker TRIPPING command	DO	1	
d	Breaker CLOSED command	DO	1	
e	Line Current (mA)	AI	1	
f	Line Voltage(VOLT)	AI	1	
g	All Relays through MODBUS TCP/IP and Multi-Function Meter through MODBUS over TCP/IP communication.			
2	Bus Coupler			
a	Breaker CLOSED Feedback	DI	1	Hardwired To PCS
b	Breaker test position	DI	1	
c	Breaker available / Trip	DI	1	
d	Breaker TRIPPING command	DO	1	
e	Bus Current	AI	1	
f	Bus Voltage	AI	1	
g	All Relays through MODBUS TCP/IP and Multi-Function Meter through MODBUS over TCP/IP Communication.			
3	Outgoing feeder – Motor feeder			
a	Breaker OFF Feedback	DI	1	Hardwired To PCS
b	Breaker ready to close	DI	1	
c	Breaker OPEN command	DO	1	
d	Breaker CLOSE command	DO	1	
4	Outgoing feeder – LV Motor Starters integral to Motor Operated Valve Open/Close			
a	Breaker OPEN Status	DI	1	Hardwired To PCS
b	Breaker CLOSE Status	DI	1	
c	Breaker OPEN command	DO	1	
d	Breaker CLOSE command	DO	1	
5	Outgoing feeder – LV Motor Starters integral to Motor Operated Valve			

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S. No.	Description Feedback	Type of I/O	No of I/O	Remarks
	Open/Close Jogging			
	Breaker OPEN Status	DI	1	Hardwired To PCS
	Breaker CLOSE Status	DI	1	
	Breaker OPEN command	DO	1	
	Breaker CLOSE command	DO	1	
	Valve Position (% Open)	AI	1	

203.28 Packing & delivery

The Vendor shall arrange seaworthy packing for all equipment. No components shall be transported as a loss items apart from Spares and Special tools & tackles.

203.29 Technical Parameters



Sl.No.	Description	Unit	Parameters
1.0	LV Switchgear – General		
a)	Type		Metal clad, free standing
b)	Type of construction		
	PCC/MCC		Form 3b
	PDB/DB		Form-2b
c)	Type of cooling		Natural air cooled
d)	Material of Construction		CRCA sheet
e)	Thickness		1.6mm
f)	Enclosure paint shade		RAL 7032 C3H
g)	Cable entry		Bottom
h)	Bus bar material - Phase & neutral (half of phase bus bar) - All vertical bus bars for multi-tier panel - Auxiliary power and control bus bars - Earth		Copper Copper Copper Copper
i)	Bus bar clearance in air (minimum) - Phase to phase - Phase to earth	mm mm	25 20
j)	Rated AC voltage	V	400 V $\pm$ 10%, 3 phase, 4 wire, Solidly earthed system



Sl.No.	Description	Unit	Parameters
k)	Insulation Voltage	V	1000
l)	Frequency	Hz	50 +3 to -5%
m)	Rated DC voltage	V	125 +10% to -15%
2.0	PCC/ MCC (400A & above) transformer incomers		
a)	One minute power frequency withstand voltage - Main circuit - Auxiliary circuit	kV kV	2.2 2.0
b)	Short circuit withstand current rating for all switchboards		50kA (rms) for 1 sec & 105kA (peak)
c)	Degree of protection of switchboards with modules in service and all doors closed		IP-42
d)	Type of Construction		
	PCC/MCCs		For ACB: Single front / Draw out Type Others: Double front/fixed type
	PDB/DBs		Single front/Double front / fixed type
3.0	Power Distribution Board (PDB/DBs) (400A & below)		
3.1	AC Distribution Board		
a)	Rated voltage	V	400 V $\pm 10\%$, 3 phase, 4 wire
b)	One minute power frequency withstand voltage - Main circuit - Auxiliary circuit	kV kV	2.2 2.0
c)	Short circuit withstand current rating	kA	50kA (rms) for 1 sec 105kA (peak)
d)	Degree of protection		IP-42
e)	Execution		Single front, fixed type.
5.0	Air Circuit Breaker		
a)	Type		ACB

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Sl.No.	Description	Unit	Parameters
b)	No. of poles		TPN/TP/4P for PCC TP/TPN for MCC
c)	Short circuit performance category		P2
d)	Rated operating duty		O-0.3s-CO-3min-CO
e)	Short time current for 1 sec.	kA(rms)	50
f)	Rated short circuit breaking current	kA(rms)	50
g)	Rated short circuit making current	kA(peak)	105
h)	Control supply voltage - Closing coil, spring charging motor - Tripping coil (2 nos)	V	125 V DC, -15%, +10% 125 V DC, -15%, +10%
7.0	Isolating Switch		
a)	Utilization category for main contacts		AC23
b)	Minimum thermal rating	A	32
8.0	Power Contactor		
a)	Permissible supply voltage variation with respect to nominal coil voltage		85% - 110%
b)	Utilization category		AC3 in general & AC4 for bi-directional motors.
c)	Type of contactor		Electromagnetic
d)	Contactor rated voltage		
e)	Contactor maximum drop out accepted		75%
f)	Nos. of Poles		3 pole
9.0	Bus Bar		
a)	Temperature rise over design ambient of 40°C (i) Bus bars		As per IEC 60947
b)	Short-time current for PCC/MCC/ACDB	kA	50kA (rms) for 1sec with 105kA (peak) dynamic rating
10.0	Auxiliary Supply		
a)	Transducer supply	V	125 DC
b)	Space heater supply	V	230 AC

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Sl.No.	Description	Unit	Parameters
11.0	Control Supply		
a)	For motor feeders rated less than 75 kW (Contractor controlled)	V	230 AC
b)	For motor feeders rated 75 kW and < 200kW (ACB controlled)	V	125 DC
c)	For ACB feeders	V	125 DC
12.0	Local Push Button Station		
a)	Contacts rating		5A at 230V AC & 2A (inductive) at 125V DC
b)	No. of contacts	Nos.	2 NO + 2 NC
c)	Degree of protection		IP-55W for outdoor IP 42 for Indoor
13.0	Local Motor Starter/Wall mounted DB		
a)	Rated voltage	V	400 V $\pm$ 10%, 3 phase
b)	Enclosure material	mm	At least 1.6 mm thick CRCA
c)	Degree of protection		IP-55
14.0	MCCB		
a)	Make		
b)	Type		
c)	Rating	A	
d)	No. of Auxiliary contacts set		
e)	Short circuit current rating	kA	
f)	Shunt trip release provided		
14.0	MPCB		
a)	Make		
b)	Type		
c)	Rated Voltage	V	
d)	Current rating	A	
e)	Short circuit current rating	kA	
15.0	MCB		
a)	Make		

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Sl.No.	Description	Unit	Parameters
b)	Type		
c)	Current Rating	A	
d)	Rated Voltage	V	
e)	Short circuit current rating	kA	
16.0	Control Transformers		
a)	Make		
b)	Type		
c)	Insulation class		
d)	kVA rating		
e)	Voltage ratio		
f)	Taps range		
17.0	Voltage Transformer		
a)	Make		
b)	Type		
c)	Reference standard		
d)	Accuracy class		
e)	Over voltage factor		
	i. Continuous		
	ii. 30 seconds		
f)	Class of insulation		
g)	1 min. power frequency voltage withstand level	kV(rms)	
h)	Basic impulse level	kV(peak)	
18.0	Internal wiring		
a)	Voltage grade	V	600/1000
b)	Colour Code		
c)	Wire Size		
d)	Conductor		Copper